

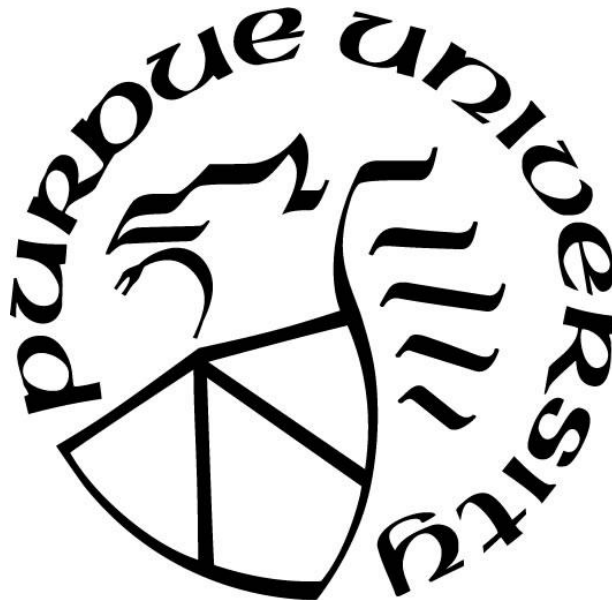
**FOREST ABOVEGROUND CARBON STOCKS IN INDIANA:
RESPONSES TO MANAGEMENT AND LIDAR-BASED ESTIMATION**

by
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A Thesis

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LIST OF ABBREVIATIONS

Abbreviation	Alias
AGB	Aboveground biomass
ANOVA	Analysis of Variance
CHM	Canopy Height Model
df	Degree of Freedom
DSM	Digital Surface Model
DTM	Digital Terrain Model
EA	Even-aged management
FACS	Forest Aboveground Carbon Stock
GPS	Global Positioning System
HEE	Hardwood Ecosystem Experiment
LAS	LASer file format
LiDAR	Light Detection and Ranging
M (or EMM)	Estimated Marginal Mean
MMYSF	Morgan Monroe and Yellowwood State Forest
NH	No harvest (control)
RFE	Recursive Feature Elimination
RMSE	Root Mean Square Error
SE	Standard Error
UEA	Uneven-aged management

ABSTRACT

Forest ecosystems play a pivotal role in climate change mitigation. Sustainable forest management practices necessitate accurate quantification of forest aboveground carbon stocks (FACS). In the first part of this study, I compared the 13-year changes in FACS across three silvicultural systems, including even-aged management (EA), uneven-aged management (UEA), and non-harvested controls (NH), in Indiana's hardwood forests. Forest stands within each silvicultural system were assigned with one of the six treatment types, including clearcutting, shelterwood, or prescribed burning for EA, single-tree selection or patch cutting for UEA, or untreated controls. From 2008 to 2021, the FACS of the study area exhibited an increase from 91.5 ± 9.0 Mg/ha to 115.3 ± 2.1 Mg/ha. Single-tree selection, shelterwood, and prescribed burning were found to have minimal impacts on FACS. However, clearcutting and patch cutting resulted in a significant reduction in FACS, with subsequent recovery reaching only 30-37% of their pre-treatment levels after 13 years. Further investigations may use long-term inventory data to analyze the chronic recovery patterns on these sites.

In the second part of this study, I evaluated the feasibility of using 3DEP LiDAR in conjunction with the random forest algorithm for multiscale FACS prediction. It was found that the stand-scale model outperformed the plot-scale model, primarily due to a stand's higher positioning accuracy and reduced boundary effects than the plot-scale model. This led to a reduction in RMSE from 25.43 Mg/ha (26%) to 16.74 Mg/ha (20%). Moreover, the stand-scale model exhibited robust landscape-level prediction performance even in scenarios where point density decreased from 7.7 points/m² to 2.0 points/m². However, the partitioned model including solely clearcut and patch sites produced a higher RMSE of 59% (17.82 Mg/ha) due to inaccurate LiDAR return classification and biased canopy height metrics extraction. Future research should delve into the mechanisms of point cloud classification to improve the FACS prediction accuracy for clearcut forest monitoring.

Overall, this thesis contributed to a deeper understanding of carbon dynamics in managed hardwood forests, highlighted the potential of using LiDAR technology for improved landscape-level carbon monitoring, and informed the decision-making processes in the context of climate change mitigation.

Keywords: Aboveground carbon stock, aboveground biomass, boundary effect, clearcut, co-registration error, even-aged management, disturbance, Indiana forest, LiDAR, point density, random forest, spatial scale, uneven-aged management

CHAPTER 1. NEAR-TERM FOREST ABOVEGROUND CARBON RESPONSES TO SILVICULTURAL TREATMENTS

1.1 Abstract

Forest carbon sequestration is being increasingly integrated into forest management practices for climate change mitigation. This research evaluated the near-term impact of even-aged (EA), uneven-aged (UEA), and no harvest (NH; control) management on forest aboveground carbon stocks (FACS) in an Indiana hardwood forest over a span of 13 years. Vegetation monitoring plots from the Hardwood Ecosystem Experiment (HEE) were categorized into six silvicultural treatments: clearcut, shelterwood, and prescribed burning for EA management; single-tree selection and patch cut for UEA management, and control or non-harvested (NH). Pre-treatment vegetation inventory data were collected during 2008-2009, followed by immediate overstory harvests. Post-treatment inventory data were collected every four years thereafter: 2013, 2017, and 2021. The live aboveground biomass of each tree was determined using allometric equations and aggregated to plot-level FACS estimates. Over 13 years, FACS at control plots increased from 91.5 ± 9.0 Mg/ha in 2008 to 115.3 ± 2.1 Mg/ha in 2021. Contrary to some studies suggesting greater carbon storage in unmanaged forests than harvested forests, results from HEE supported an alternative argument that intermediate management retaining canopy structure had minimal impact on FACS. Specifically, single-tree selection and prescribed burning resulted in FACS closest to untreated controls throughout all post-treatment measurement periods. For three-staged shelterwood, the first two cuts combined initially led to a lower FACS estimates than the concurrent level in NH stands, but FACS returned to pre-treatment level within a decade. Lastly, intensive management including clearcutting and patch cutting resulted in the lowest FACS among all treatments, with 30-37% recovery over 13 years. At present, the untreated forest within HEE has lower aboveground carbon stocks but a higher annual carbon accumulation rate than less disturbed mature forests in the central hardwood region. Like many other hardwood forests in the eastern US, many forests in Indiana have yet to fully recover from historical land clearing and subsequent abandonment in the 1920s. Rapid carbon accumulation reinforced the significant role of these middle-aged, secondary forests in addressing climate change. In conclusion, this research provided new evidence for similar carbon stock evaluations and for future studies investigating long-term carbon effects of various silvicultural prescriptions.

Keywords: aboveground biomass, carbon stocks, even-aged management, clearcut, shelterwood, hardwood forests, silviculture, single-tree selection, stand age, uneven-aged management

1.2 Introduction

Forest ecosystems are a crucial natural climate solution for offsetting greenhouse gas emissions from human activities. Through photosynthesis, forests absorb carbon dioxide from the atmosphere and convert it into carbohydrates. In 2021, the net uptake of carbon by forest ecosystems in the United States was approximately 161.6 million metric tons, accounting for 11% of the country's total greenhouse gas emissions (EPA, 2023). With climate change increasingly influencing forest productivity, there is growing global attention on enhancing carbon sequestration potential through sustainable forest management (Ameray et al., 2021; Reyer et al., 2014). Quantifying forest carbon sequestration potential often involves considering both carbon stocks and rates of change (Hoover & Smith, 2023). Carbon stocks denote the absolute amount of carbon stored in the forest at a given time, while rates of change reflect carbon stock exchange over a specified period (Hoover & Smith, 2021). Both metrics are crucial in forest management planning to effectively achieve carbon objectives, as forests with high carbon stocks may not necessarily exhibit high carbon sequestration rates. Forest stand age significantly influences carbon dynamics. Young forests generally store less carbon than mature forests due to simpler canopy structure, but the carbon accumulation rate in younger stands is usually higher than older stands (Birdsey et al., 2023a; Hoover & Smith 2023; Pregitzer & Euskirchen, 2004; Watts et al 2020). Nonetheless, forest carbon dynamics are often a function of multiple factors, including structural complexity, species composition, stand age, site productivity, and management or disturbance history (Anderson et al., 2023; Conner et al., 2015; D'Amato et al. 2011; Fraser et al., 2023; Ford & Keeton 2017; Hoover & Smith 2023; Puhlick et al., 2019; Urbano & Keeton 2017).

Disturbance plays a significant role in altering forest canopy structure and driving carbon stock change. It may lead to a reduction in carbon stocks, but the subsequent regrowth of forests acts as a carbon sink (Birdsey et al., 2006; Pugh et al., 2019; Williams et al., 2016). The dominant disturbance type in the eastern US is harvesting, contributing to 1.6% of total non-soil carbon stock removal over a 30-year period from 1990 to 2011 (Birdsey et al., 2019). Conversely, other disturbances, such as wind and fire, only accounted for 0.1-0.2% of carbon stock removal.

Therefore, understanding of how harvesting practices reshape forest carbon dynamics provides valuable insights for public agencies and private forestland owners to develop climate-adaptive forest management practices.

The Central Hardwood Region (CHR) is an important carbon sink in North America, spanning from Massachusetts to Missouri and supporting 39% of the total growing stock in the continental US (Hicks, 1998; Jin et al., 2017; Powell et al., 1993). However, historical land clearing and intensive timber harvesting during the late 19th and early 20th centuries have left most forests in the CHR relatively young, with an average stand age ranging between 40 to 90 years (Jin et al., 2017; Pan et al., 2011; Shifley et al., 2014). Morgan-Monroe and Yellowwood State Forests (MMYSF), located in south-central Indiana, are typical representations of forests in the CHR that have recovered from extensive land clearing and timber harvesting, and are characterized by oak-hickory communities. However, the lack of even-aged management and fire suppression since the 1920s have not only inhibited oak regeneration but also favored the growth of shade-tolerant species, such as maple (*Acer* spp.) and beech (*Fagus grandifolia*). This succession-driven species replacement, termed mesophication, is common throughout eastern North America and has been reported by many authors (Lowney et al., 2015; Spetich & Parker 1998, Ma et al., 2016; Morrissey et al 2015; Nowacki and Abrams 2008; Thompson et al., 2013). The transition towards shade-tolerant species has raised concerns among ecologists and managers, as oak-hickory forests offer both economic and ecological values, such as timber production and serving as vital food sources for wildlife. The Hardwood Ecosystem Experiment (HEE) is a 100-year multi-disciplinary project that studies the ecological and human aspects of forest management in MMYSF. One of HEE's primary goals is to develop silvicultural prescriptions that maintain oak-dominated forest landscapes in southern Indiana (Kalb & Mycroft, 2013). Previous publications from HEE have primarily focused on the ecological impact of forest management on plant and wildlife communities, and there is lack of research on how different silvicultural systems influence forest carbon dynamics. Debates exist on how different silvicultural treatments influence existing forest aboveground carbon stocks (FACS) and how FACS recover post-treatment (Ameray et al., 2021; Anderson et al., 2023; Chen et al., 2015; Conner et al., 2015; Davis et al., 2009; Ford & Keeton, 2017; Thom & Keeton, 2020). This study aims to assess and compare pre- and post-treatment forest aboveground carbon stocks (FACSs) over a 13-year period across three silvicultural systems at HEE: even-aged (EA), uneven-aged (UEA), and non-harvested (NH) control. Within each

silvicultural system, forest stands were assigned with one of the six treatment types, including clearcutting, shelterwood, or prescribed burning for EA, single-tree selection or patch cutting for UEA, or untreated controls. I predicted that the highest FACSs would be found in untreated control forests, as no harvesting was implemented throughout all inventory periods. Conversely, I predicted that the clearcut forests would have the lowest FACSs because this harvesting method involves entire overstory removal.

1.3 Methods

1.3.1 Study area

Located in south-central Indiana (+39.25, -86.39; **Figure 1-1**), the MMYSF is approximately 20000 hectares in size. The MMYSF remained uncovered by the Wisconsin glaciation and is characterized by rugged terrains with silt-loam soils (Jenkins, 2013). Elevation at MMYSF ranges from 167 m to 300 m, with an average slope of 18.5 degrees (Barnes et al., 2016; Shao et al., 2018). July mean temperature is 24 °C, January mean temperature is -1 °C, and the average annual precipitation is 1274 mm. As of 2013, over 50% of forests within MMYSF were over 80 years old and only 2% were younger than 20 years old (Haulton, 2013). A blend of tree species is found across MMYSF, which typically includes black oak (*Quercus velutina*), white oak (*Q. alba*), northern red oak (*Q. rubra*), pignut hickory (*C. glabra*), and sugar maple (*Acer saccharum*) (Saunders & Arseneault, 2013). Other species such as tulip poplar (*Liriodendron tulipifera*) and sassafras (*Sassafras albidum*) also occur across the southeast forests of MMYSF (Kalb & Mycroft, 2013). Oak-hickory forest types represent 43% of the total property area (Haulton, 2013). Since MMYSF was established during the mid-20th century, single-tree selection and small-group selection have been the primary silvicultural methods used for this regime to thin overstocked stands and improve tree quality (Kalb & Mycroft, 2013). The most recent timber harvesting prior to HEE occurred during 1971-2006, which removed an average of 27% volume from each research core (Haulton, 2013).

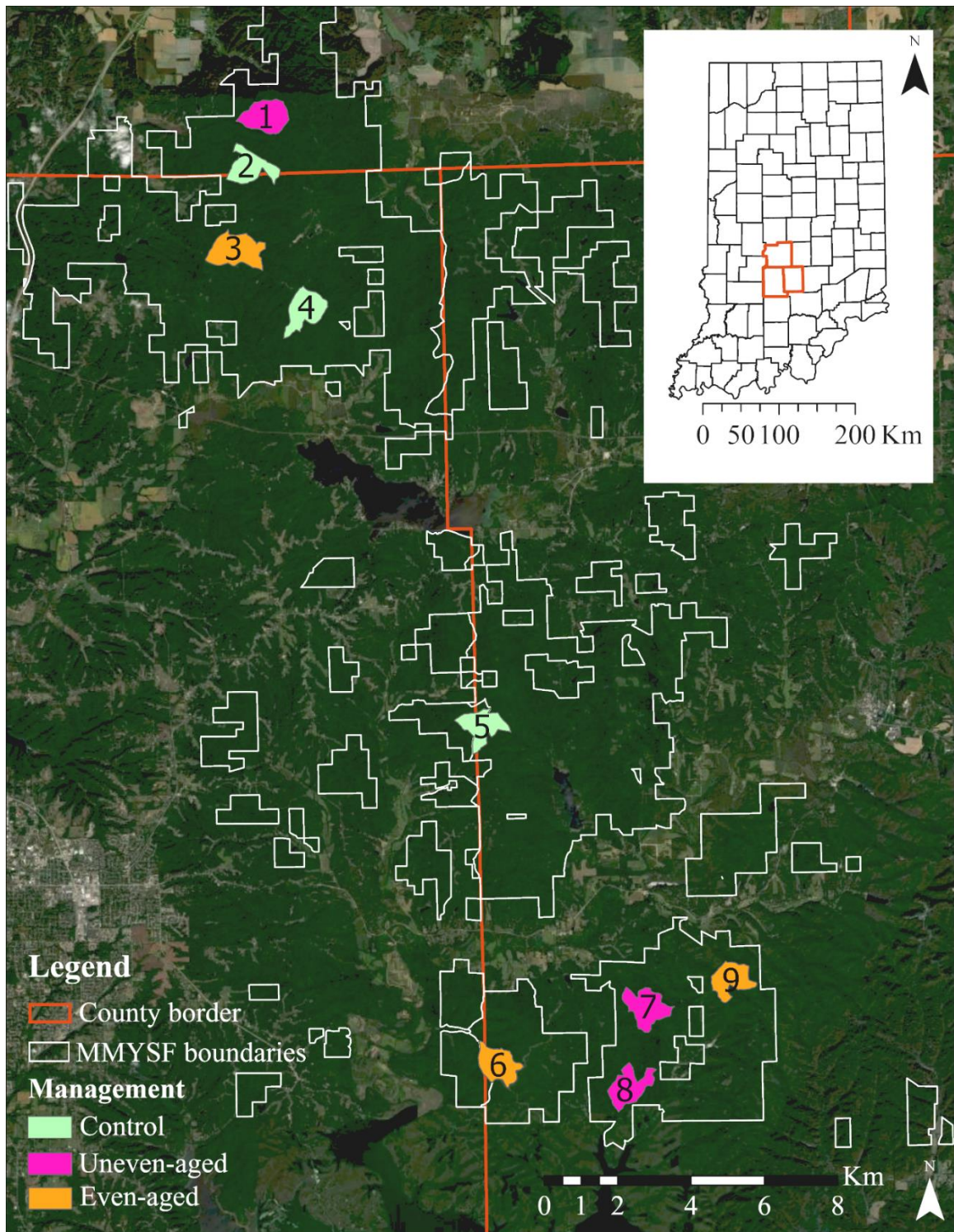


Figure 1-1. Map of research cores for the Hardwood Ecosystem Experiment (HEE) in Indiana. Cores # 2, 4, 5 received no management (control); cores # 1, 7, 8 received uneven-aged management; cores # 3, 6, 9 received even-aged management. Map is derived from Kalb & Mycroft (2013) with minor changes.

1.3.2 Experimental design

Vegetation inventory data were collected by staff and scientists managing the HEE. Nine research units were distributed throughout HEE sites (**Figure 1-1**), with each unit consisting of a research core and a buffer area. The total area of HEE was 3603 ha, equivalent to about 20% of the entire MMYSF landscape (Kalb & Mycroft, 2013). Each research core, 80-100 hectares in size, was randomly assigned one of the three management approaches: 1) even-aged management (EA); 2) uneven-aged management (UEA); and 3) unmanaged control (NH).

Table 1-1. Summary of treatment stands and area (hectares) across all research cores at HEE.

Management	Core	Treatment	Total stands	Mean area	Sd	Min	Max	Total area
Control (NH)	2,4,5	NA	3	86.1	3.4	83.5	89.9	258.2
Uneven-aged (UEA)	1,7,8	Single-tree	3	80.4	4.0	75.9	83.3	241.1
		Patch Cut	24	1.1	0.7	0.2	2.6	27.5
Even-aged (EA)	3,6,9	Clearcut	6	3.9	0.6	2.7	4.4	23.5
		Shelterwood	6	4.7	0.5	4.1	5.5	28.5
		Burning	12	4.8	0.9	3.3	5.5	57.1
		EA Untreated*	37	4.5	1.1	2.6	6.5	165.3

Note: * represents EA stands that have not received any treatment by the end of the 2021 inventory. Sd = standard deviation.

Up to 2021, each EA core included two clearcut stands, two shelterwood stands, and four prescribed burning stands (**Table 1-1**). Additionally, there were 37 stands in total from all even-aged cores left unmanaged for future treatments. Each even-aged stand was approximately 4.05 ha in size. Harvest boundaries followed topographic features (e.g., ridgetops, gullies) to best capture slope variations. For clearcut harvest, all trees > 30.48 cm DBH (diameter at breast height; 1.37m) were removed during the overstory harvest in 2008-2009; all remaining woody stems were either felled, coppiced to within 15.24 cm from the ground, or chemically treated (Kalb & Mycroft 2013). The three-stage shelterwood treatment had its preparatory cut during 2008-2009, received an establishment harvest during 2015-2016 to reduce overstory stocking, and will receive a final regeneration harvest in 2028. The preparatory cut removed the understory and midstory non-oak species that were less than 25.4 cm dbh, but avoided creating large canopy gaps and maintained the basal area targeting at 13.8 m²/ha. During 2015-2016, the establishment cut reduced overstory

stocking to 13.8-16.1 m²/ha basal area by removing poorly formed overstory trees to favor growth of vigorous large-crowned oak-hickory species (Kalb & Mycroft, 2013). The prescribed burning treatment focused on midstory and understory removal to reduce fuel accumulation, litter, and duff depth to favor oak-hickory seedling establishment (Stanis & Saunders, 2018). The prescribed burning stands will serve as the future locations for clearcut or shelterwood treatments, to be harvested during 2028-2029. To reduce understory and midstory densities and improve oak regeneration, two burns occurred during 2015-2017 and 2020-2023, respectively. For UEA management, each core included one single-tree selection stand and eight patch cut stands. Single-tree selection was executed for 90% of the area within each uneven-aged core, targeting a 16.1-22.9 m²/ha basal area while retaining trees within a 15.2 m buffer from patch cut sites. Patch cuts were distributed within single-tree treatments; the boundaries of patch cuts were 4-sided polygons and were placed to best capture slope variations. The actual harvest size of patch cuts ranged from 0.15 ha to 2.56 ha. For control, research cores 2, 4, and 5 received no silvicultural treatment. As control cores have no harvest boundaries, I considered each core a stand (**Table 1-1**).

Right before the overstory harvest during 2008-2009, a pre-treatment vegetation inventory was conducted to obtain baseline information from each research core (Kalb & Mycroft, 2013), (**Figure 1-2**). Post-treatment vegetation inventories were conducted at four-year intervals: 2013-2014, 2017-2018, and 2021-2022. For this study, I simply refer to the four inventory periods as “2008”, “2013”, “2017”, and “2021”. Each research core included approximately 80 systematically distributed, permanent sample plots. Due to limited effort during the pre-treatment inventory, the sampling density was especially low for plots in control units and untreated plots in even-aged units, but it increased over time and stabilized since the 2017 inventory. In 2021, the average sampling density across all treatments ranged from 8% to 12% (**Table 1-2**). Each circular sample plot was 0.1 ha in size with a 75 m west-east spacing and a 150 m north-south spacing. Within each plot, all overstory trees greater than 11.5 cm DBH and saplings of 1.5 - 11.5 cm DBH were measured by: DBH, species, crown class, azimuth and distance to the plot center, living condition, stratum/light field, and height. Tree height was measured for only about 20% of total inventoried stems. In 2013, stand boundaries of clearcuts patch cuts were re-measured with a GPS device with an average accuracy of 5 m; stand boundaries of other treatments were hand-digitized in ArcGIS with the aid of topographic maps. As plots were established prior to the harvests, I assigned a treatment attribute to each plot based on the treatment stand it falls within. For plots

located between two different treatment stands, the treatment attribute was determined by the majority rule.

Table 1-2. Vegetation sampling plot counts per treatment during 2008-2021. Sampling density was calculated as the number of plots in 2021 divided by total treatment area.

Management	Treatment	Inventory year				Sampling density (2021)
		2008	2013	2017	2021	
Control	Control	13	118	225	225	9%
Uneven-aged (UEA)	Single-tree	200	139	201	201	8%
	Patch Cut	33	17	32	33	12%
Even-aged (EA)	Clearcut	23	22	23	23	10%
	Shelterwood	15	24	24	24	8%
	Burning	15	44	45	44	8%
	EA Untreated*	52	155	155	155	9%

Note: * represents even-aged stands that have not received any treatment by the end of the 2021 inventory and they were excluded from FACS calculation.

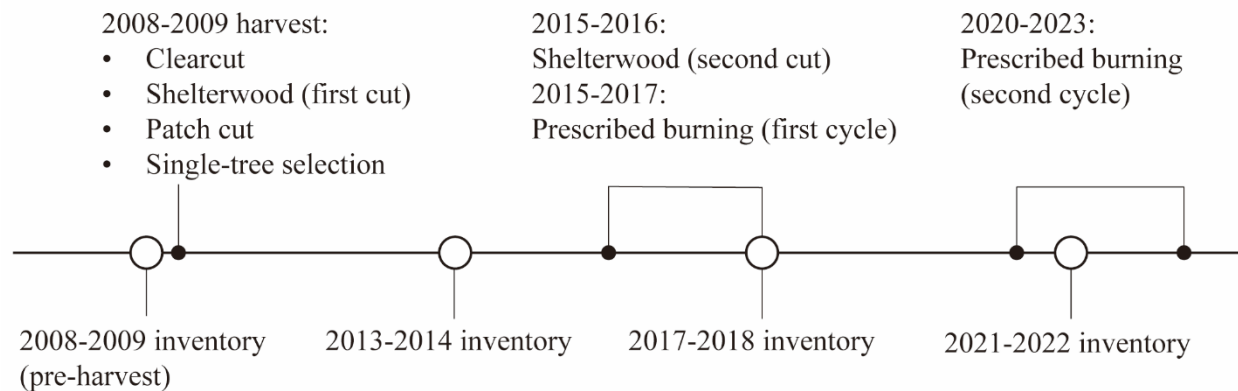


Figure 1-2. Timeline of vegetation inventories and silvicultural treatments at HEE from 2008 to 2023. Inventory and treatment years are represented by blank circles and black dots, respectively.

1.3.3 Data analysis

Inventory data were analyzed in R version 4.3.1 (R Core Team, 2023). To evaluate live aboveground carbon dynamics from each treatment during each inventory period, DBH-based allometric equations were first used to calculate AGB for each live tree (Chojnacky et al. 2013) and then scaled to plot-level AGB density, expressed in units of metric tons per hectare (Mg/ha).

The equations used defined AGB as all material aboveground and were based on taxonomic groupings with different wood specific gravities (Jenkins et al., 2003, Jenkins et al., 2004). As most equations required a minimum DBH at 3 cm, inventoried stems with DBH less than this threshold were excluded from AGB calculation. The plot-level AGB was then multiplied by 0.5 to obtain forest aboveground carbon stocks (FACS) (Hoover & Stout, 2007; Maclean et al., 2014; Anderson et al., 2023).

Due to lack of samples during 2008-2013, repeated-measures ANOVA was not robust for site-specific pre- and post-treatment carbon comparisons. Instead, I used one-way ANOVA to assess if there was any significant difference in FACS among treatments per inventory period. For each inventory year, a linear model was created to fit FACS over treatments. In total, four linear models were created and all of them satisfied residual normal distribution and homogeneity of variance. In the case of significant results from ANOVA ($\alpha = 0.05$), post-hoc Tukey's pairwise comparison was conducted to determine the probability of differences among each pair of treatment levels. For each treatment, the estimated marginal mean (EMM) and the standard error (SE) were calculated via the "emmeans" package. To quantify FACS change over time within each treatment, I calculated both the absolute and relative change between the most recent post-treatment inventory period (2021) and the pre-treatment period (2008). The relative change was defined as the absolute change of FACS divided by the FACS at the pre-treatment stage, expressed as a percentage value. A positive value indicated the current FACSs have exceeded the pre-treatment level, whereas a negative value indicated the FACS lower than the pre-treatment level. To evaluate trends in post-harvesting FACS accumulation, mean annual increment (MAI) was calculated as the absolute change during 2013-2021 divided by eight years. There was an exception in calculating the above three metrics for the burning plots. As the first burning cycle started in 2015, I considered the 2013 measurement as its pre-treatment inventory. In other words, the duration used to calculate absolute and relative difference between the pre-treatment and the latest post-treatment inventory was eight years rather than 13. Similarly, the duration used to calculate MAI was four years instead of eight. Since the estimation of absolute change, relative change, and MAI were not directly derived from paired samples but EMMs at two points in time, I used formulas from error propagation to calculate the standard errors for the last three metrics, considering that the EMMs were created for the purpose of among-treatment comparisons in a single year rather than within-treatment comparisons across multiple years.

$$SE_{Abs.change} = \sqrt{SE_{EMM_{t1}}^2 + SE_{EMM_{t2}}^2} \quad (1)$$

$$SE_{Rel.change} = \left| \frac{EMM_{t2}}{EMM_{t1}} \right| * \sqrt{\left(\frac{SE_{EMM_{t1}}}{EMM_{t1}} \right)^2 + \left(\frac{SE_{EMM_{t2}}}{EMM_{t2}} \right)^2} \quad (2)$$

$$SE_{MAI} = \frac{SE_{Abs.change}}{|t_2 - t_1|} \quad (3)$$

Where:

SE = standard error (in Mg/ha for (1), (3) and % for (2));

Abs. or Rel. change = absolute or relative change in Mg/ha and %, respectively;

EMM_{t1} or EMM_{t2} = estimated marginal mean (Mg/ha) during the pre-treatment period and the most recent inventory period, respectively;

$t_2 - t_1$ = the number of years between the first and the last post-treatment period.

1.4 Results & Discussion

The pre-treatment inventory in 2008 showed no across-site difference in FACS (F (5, 293) = 0.95, $p = 0.449$; **Table 1-4, Table 1-3, Figure 1-3**), ranging between 88.6 ± 6.8 Mg/ha to 104.4 ± 5.7 Mg/ha. However, significant differences were found in all post-treatment inventories in 2013 (F (5, 358) = 51.33, $p < 0.001$), 2017 (F (5,544) = 71.03, $p < 0.001$), and 2021 (F (5,544) = 62.19, $p < 0.001$).

1.4.1 FACS responses to no management

Over 13 years, FACSs in untreated control plots have consistently increased from 91.5 ± 9.0 Mg/ha to 115.3 ± 2.1 Mg/ha (**Table 1-3, Figure 1-3**). It was surprising to observe a $26\% \pm 12.6\%$ (23.8 ± 9.3 Mg/ha) increase of FACS on control plots over 13 years of time (MAI = 1.2 ± 0.4 Mg/ha/yr), which was the highest among all treatment types. As only 15 plots were available to assess the pre-treatment conditions in control units, estimates of FACS change inevitably included a large standard error. However, even if assuming minimal change during 2008-2021, the rate of increase was still higher than the average rate reported for old-growth forests in the CHR

(7% or 9.2 ± 1.5 Mg/ha over a 20-year period, MAI = 0.5 ± 0.1 Mg/ha/yr; Fraser et al., 2023). This was probably due to younger stand ages at HEE as opposed to other conventionally defined old-growth forests in CHR. Without severe environmental perturbations, younger stands typically accrue AGB and carbon stocks more rapidly than older stands (Pugh et al 2019). As of 2021, the stand age of unmanaged forests at HEE was approximately 75 to 110 years (Haulton, personal communication). In contrast, old-growth forests in Indiana had a stand age between 130-300 years (Fraser et al., 2023; Lowney et al., 2016). As younger stands typically had a higher carbon sequestration potential than older stands (Birdsey et al., 2023a; Hoover & Smith, 2003; Jin et al., 2017; Nunery & Keeton, 2010; Pan et al., 2011), a higher MAI was expected from HEE. Furthermore, although the FACS of unmanaged forests at HEE was lower than those from disturbed old-growth forests in the CHR (125 to 134 Mg/ha), it was comparable to secondary forests of similar stand age in other parts of the eastern US. The live tree carbon stocks for a 60-to-100-year temperate hardwood forest typically ranged between 100-150 Mg/ha. Anderson et al (2023a) reported 100 Mg/ha FACS for an 80-year-old oak stand in Missouri Ozarks. Ford & Keeton (2017) reported 113 Mg/ha FACS for a 70-to-100-year-old hardwood-conifer forest in northeast US. In southeast New York state, a 90-to-115-year-old oak-dominant stand stored 95 Mg/ha FACS on average (Schuster et al., 2008). Although secondary forests had less carbon stocks than old-growth forests, they occupy 75% of the forest area in the eastern US whereas old-growth forests only comprise 5% of forest area (Anderson et al., 2023; Birdsey et al., 2023a). The wide distribution of secondary forests in the eastern US is largely attributed to the historical land clearing and subsequent land abandonment during the early 20th century, leaving an average stand age of 50-90 years in the present day (Birdsey et al., 2023b; Jin et al., 2017; Pan et al., 2011). Rapid accumulation of FACS from untreated control plots provided additional evidence that the middle-aged, secondary forests are an important carbon sink in the eastern US.

Table 1-3. Summary of estimated marginal means (EMM) and standard errors of FACS (Mg/ha) by treatment types from nine research cores at HEE. Abs. change, percent relative change (% rel. change), and MAI denote absolute change, percent relative change, and mean annual increment, respectively.

Treatment sites	Inventory year				Abs. change	% Rel. change	MAI
	2008	2013	2017	2021			
Control	91.5	106.1	108.7	115.3	23.8	26.0	1.2
	(9.0)	(2.7)	(2.0)	(2.1)	(9.3)	(12.6)	(0.4)
Single-tree	98.7	99.3	101.7	107.6	8.9	9.0	1.0
	(2.3)	(2.5)	(2.1)	(2.2)	(3.2)	(3.4)	(0.4)
Patch Cut	104.4	39.1	31.6	38.4	-66.0	-63.2	-0.1
	(5.7)	(7.1)	(5.3)	(5.5)	(7.9)	(5.7)	(1.1)
Clearcut	88.6	14.2	18.7	26.9	-61.7	-69.6	1.6
	(6.8)	(6.2)	(6.3)	(6.6)	(9.5)	(7.8)	(1.1)
Shelterwood	93.3	96.6	88.7	91.9	-1.4	-1.5	-0.6
	(8.4)	(6.0)	(6.1)	(6.5)	(10.6)	(11.3)	(1.1)
Burning	91.0	106.1	107.5	111.2	5.1*	4.8*	0.9*
	(8.4)	(4.4)	(4.5)	(4.8)	(6.5)	(6.3)	(1.6)

*: As the first burn occurred during 2015-2017, the duration used to calculate absolute and relative change was 8 years (2013-2021) instead of 13 (2008-2021) which were used for all other treatments (see methods). Similarly, the duration used to calculate post-burning MAI was 4 years (2017-2021) instead of 8 (2013-2021).

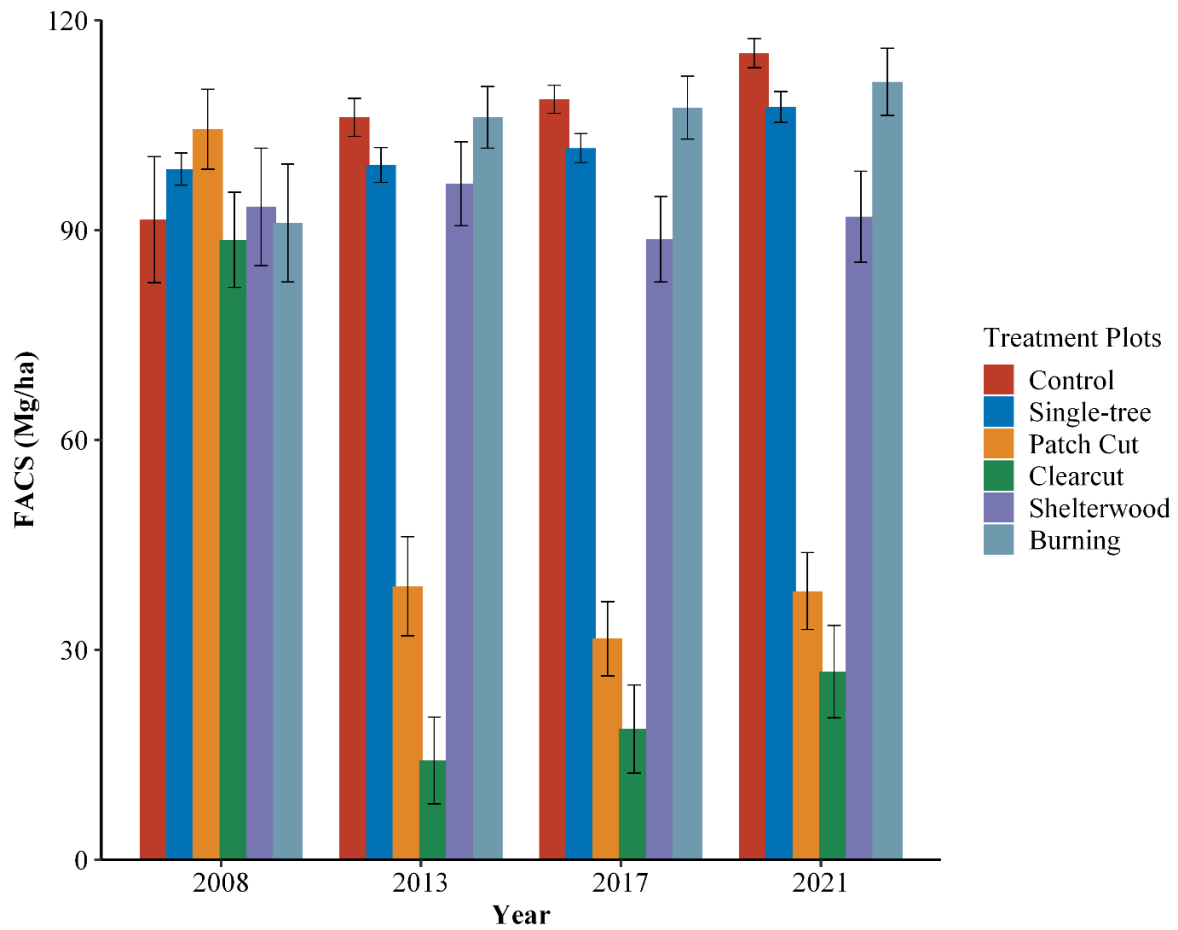


Figure 1-3. Estimated marginal means of aboveground carbon stocks (FACS) and standard errors across treatment plots during 2008-2021. Treatments from left to right include control (red), single-tree selection (blue), patch cut (orange), clearcut (green), shelterwood (purple), and prescribed burning (livid).

Table 1-4. Summary of ANOVA of treatments from 2008 to 2021.

	Df	Sum of squares	Mean square	F-value	P-value	Year
Treatment	5	5026	1005.1	0.95	0.449	2008
Residuals	293	310200	1058.9			
Total	298	315226				
Treatment	5	219300	43855.1	51.33	<0.001	2013
Residuals	358	305800	854.3			
Total	363	525100				
Treatment	5	319600	63918.6	71.03	<0.001	2017
Residuals	544	489500	899.9			
Total	549	809100				
Treatment	5	314900	62977.4	62.19	<0.001	2021
Residuals	544	550900	1012.7			
Total	549	865800				

Note: Mean squares are based on type III sums of squares.

1.4.2 FACS responses to intensive management

As expected, a significant decrease of FACS was found from intensive harvesting approaches, including clearcutting in EA and patch cutting in UEA management. Both treatments resulted in the lowest FACS estimates throughout all post-treatment inventories (14.2 ± 6.2 Mg/ha to 39.1 ± 7.1 Mg/ha; **Table 3, Figure 3**). The patch cut system is essentially a type of clearcutting, but it generally creates smaller openings less than 2 hectares in size (Table 1; Kellner & Swihart, 2016; Yamasaki et al., 2014). Although the average FACS estimates in patch cuts seemed consistently higher than clearcuts across all post-harvest inventory years, pairwise comparisons found no significant difference (**Table 5**). This was probably because about 20% of the patch cut plots were located at the harvest boundary and tallied unharvested trees within the buffer zone between the patch cut and the single-tree selection treatment. Instead, plots categorized as “clearcut” did not have this bordering issue, as nearly all of them were located within the interior of the harvesting area. For this reason, the FACS estimates from clearcut plots better represent values in large openings. Over 13 years since clearcutting, the FACS have only recovered to 30% (26.9 ± 6.6 Mg/ha) of its pre-treatment level. This relative rate of recovery was comparable to regional averages in other parts of the CHR. For example, for oak-hickory forests at Missouri Ozarks,

recovery from clearcutting was 21% (30.7 Mg/ha) over 18 years (Chen et al., 2015). It was reasonable for Missouri Ozarks to have a lower recovery rate of carbon stocks than MMYSF because the site productivity in Missouri Ozarks is generally lower than that of forests in southern Indiana (Anderson et al., 2023; Fraser et al., 2023). In the southern Appalachian Mountains, however, the recovery from clearcutting was faster (40 - 67%; 41 - 48 Mg/ha) for mixed-oak communities over 20 years (Elliot et al., 2002). Compared with other treatments, clearcutting plots from the HEE had the highest mean annual growth of FACS (1.6 ± 1.1 Mg/ha/yr) among all treatments. This value was typical for forests in the central US with a stand age between 0-20 years (1.17 Mg/ha/yr; Hoover & Smith, 2023). As clearcut forests are still in their re-establishing phase, they are likely to continue maintaining a high MAI until overstory canopy closes. Then, the FACSs will continue to increase but the growth rate will likely slow and eventually level off when stand age reaches 100 years old (Birdsey et al., 2023a). This may be tested by the future inventory data.

Table 1-5. Pairwise comparison of FACSs between treatments from 2013 to 2021.

Pairs	df	T-ratio	P-value	Estimate	SE	Year
Control - Single	359	1.857	0.431	6.8	3.7	2013
Control - Patch Cut	359	8.833	<.001	67.0	7.6	2013
Control - Clearcut	359	13.548	<.001	92.0	6.8	2013
Control - Shelterwood	359	1.452	0.695	9.5	6.5	2013
Control - Burning	359	0.001	1.000	0.0	5.2	2013
Single - Patch Cut	359	8.014	<.001	60.2	7.5	2013
Single - Clearcut	359	12.699	<.001	85.2	6.7	2013
Single - Shelterwood	359	0.419	0.998	2.7	6.5	2013
Single - Burning	359	-1.342	0.761	-6.8	5.1	2013
Patch Cut - Clearcut	359	2.647	0.089	25.0	9.4	2013
Patch Cut - Shelterwood	359	-6.203	<.001	-57.5	9.3	2013
Patch Cut - Burning	359	-8.024	<.001	-67.0	8.4	2013
Clearcut - Shelterwood	359	-9.558	<.001	-82.5	8.6	2013
Clearcut - Burning	359	-12.048	<.001	-92.0	7.6	2013
Shelterwood - Burning	359	-1.280	0.796	-9.5	7.4	2013

Table 1-5 continued

Control - Single	544	2.401	0.158	7.0	2.9	2017
Control - Patch Cut	544	13.601	<.001	77.1	5.7	2017
Control - Clearcut	544	13.701	<.001	90.0	6.6	2017
Control - Shelterwood	544	3.115	0.024	20.1	6.4	2017
Control - Burning	544	0.252	1.000	1.2	4.9	2017
Single - Patch Cut	544	12.277	<.001	70.1	5.7	2017
Single - Clearcut	544	12.567	<.001	83.0	6.6	2017
Single - Shelterwood	544	2.018	0.333	13.1	6.5	2017
Single - Burning	544	-1.163	0.854	-5.8	5.0	2017
Patch Cut - Clearcut	544	1.571	0.618	12.9	8.2	2017
Patch Cut - Shelterwood	544	-7.040	<.001	-57.0	8.1	2017
Patch Cut - Burning	544	-10.934	<.001	-75.9	6.9	2017
Clearcut - Shelterwood	544	-7.987	<.001	-69.9	8.8	2017
Clearcut - Burning	544	-11.540	<.001	-88.7	7.7	2017
Shelterwood - Burning	544	-2.483	0.131	-18.8	7.6	2017
Control - Single	544	2.478	0.133	7.7	3.1	2021
Control - Patch Cut	544	12.957	<.001	76.9	5.9	2021
Control - Clearcut	544	12.686	<.001	88.4	7.0	2021
Control - Shelterwood	544	3.418	0.009	23.4	6.8	2021
Control - Burning	544	0.784	0.970	4.1	5.3	2021
Single - Patch Cut	544	11.579	<.001	69.2	6.0	2021
Single - Clearcut	544	11.523	<.001	80.7	7.0	2021
Single - Shelterwood	544	2.285	0.202	15.7	6.9	2021
Single - Burning	544	-0.668	0.985	-3.5	5.3	2021
Patch Cut - Clearcut	544	1.332	0.767	11.5	8.6	2021
Patch Cut - Shelterwood	544	-6.267	<.001	-53.5	8.5	2021
Patch Cut - Burning	544	-9.927	<.001	-72.8	7.3	2021
Clearcut - Shelterwood	544	-7.002	<.001	-65.0	9.3	2021
Clearcut - Burning	544	-10.291	<.001	-84.3	8.2	2021
Shelterwood - Burning	544	-2.383	0.164	-19.2	8.1	2021

Note: The p-value presented in this table is adjusted by the Tukey method. Df = degree of freedom; SE = standard error.

1.4.3 FACS responses to intermediate management

Plot data indicated that intermediate silvicultural treatments that retained canopy structure may not cause a significant change in FACS. Single-tree selection treatment exhibited increased FACS over 13 years (8.9 ± 3.2 Mg/ha, $9.0\% \pm 3.4\%$) with a post-treatment MAI at 1.0 ± 0.4

Mg/ha/yr. Although plots from single-tree selection treatments had consistently less FACS (98.7 ± 2.3 Mg/ha to 107.6 ± 2.2 Mg/ha) than plots of untreated controls (91.5 ± 9.0 Mg/ha to 115.3 ± 2.1 Mg/ha) by about 7 Mg/ha during each inventory period, this difference was not significant (**Table 5**). In this study, the target basal area for single-tree selection ($16.1 - 22.9$ m²/ha) was just slightly below its pre-treatment level ($21.7 - 29.9$ m²/ha; Saunders & Arseneault, 2013), suggesting that the amount of timber removal from single-tree selection was insufficient to cause a drastic reduction of carbon stocks. Compared to clearcutting, which removed the entire overstory, single-tree selection only removed a small number of overstory trees to mimic the natural gap dynamics in mature, unmanaged stands (Hicks 1998). The creation of canopy gaps enhanced the structural complexity of the forest, while the majority of aboveground biomass and carbon stocks were still stored within overstory trees in this harvest regime (Chen et al., 2015). Previous studies have demonstrated that structurally complex forests had high productivity (Ford & Keeton, 2017; Gough et al., 2019; Thom & Keeton, 2020; Wang et al., 2011). In MMYSF, the mean annual change of FACS from single-tree selection plots was 1.0 ± 0.4 Mg/ha/yr, which was very close to the untreated control plots (1.2 ± 0.4 Mg/ha/yr). Ford & Keeton (2017) also reported similarities in mean annual carbon flux between uneven-aged management (1.2 ± 0.3 Mg/ha/yr) and control (1.3 ± 0.4 Mg/ha/yr) 10 years following harvesting in the Green Mountains in the northeast US. In the Missouri Ozarks, the annual growth rate of FACS was even higher for forests with single-tree selection harvest (1.4 Mg/ha/yr) than for unharvested forests (0.9 Mg/ha/yr, Chen et al., 2015). These findings reinforce the argument that reduced-intensity harvesting has minimal impact on forest carbon sequestration rates (Brown et al., 2018; Davis et al., 2009; Jin et al., 2017).

Pairwise comparisons also found that prescribed burning had no significant impact on FACS, which was consistent to the values from untreated controls throughout all measurement periods. A similar finding was also documented for oak-hickory forests in southern Ohio (Chiang et al., 2008). This was not surprising since the intensity of prescribed fires in the Midwest are much lower than wildfires associated with stand replacement. Wildfires occurring outside of prescription conditions could lead to significant overstory tree mortality, whereas prescribed burning mostly results in understory mortality. In 2015, two stands on the HEE were burned, but all trees survived and no change in crown class or strata was found during the 2017 inventory (Stanis & Saunders, 2018), thus ensuring the stability of overall FACS after the burning. While burning does result in

the release of carbon into the atmosphere, carbon emissions from combustion were not considered in this study.

For the shelterwood treatment, the preparatory cut in 2008-2009 did not significantly affect FACS, but the establishment cut during 2015-2016 resulted in a significant reduction in FACS compared to controls ($t = -3.115$, $p = 0.024$). In 2017, the average FACS of control plots was 108.7 ± 2.0 Mg/ha whereas the shelterwood plots contained 88.7 ± 6.1 Mg/ha. Four years later in 2021, shelterwood plots (91.9 ± 6.5 Mg/ha, $t = -3.418$, $p = 0.009$) still had significantly lower FACS than the controls (115.3 ± 2.1 Mg/ha). This could be attributed to the difference in post-harvest stand structure. The preparatory cut focused on midstory and understory removal but avoided large gap formation (Kalb & Mycroft, 2013). In contrast, the establishment cut was intended to open the stand and create an evenly distributed overstory by removing poorly shaped canopy and subcanopy trees. Since overstory trees store the most carbon in a stand, the second harvest may have removed more FACS than the first one. However, it should also be noted that the “lag time” between the harvest and the following inventory was longer for the first cut than the second cut. The preparatory cut was executed right after the pre-treatment inventory during 2008-2009, but the first post-treatment inventory was conducted four years later. In comparison, the second post-treatment inventory was conducted only one year after the establishment cut during 2015-2016. This suggested an alternative possibility that the preparatory may have caused a loss of carbon stocks, but this reduction was not shown from the 2013 data because it was offset by the regrowth of forests over four years. Nonetheless, both harvests were far less than intense clearcutting. The most important finding for the three-staged shelterwood treatment was that the loss of FACS from the first two cuttings combined were offset within 13 years by forest regrowth (-1.4 ± 10.6 Mg/ha, $-1.5\% \pm 0.2\%$, $MAI = -0.1 \pm 0.8$ Mg/ha; Table 3). This indicated that the future estimation of FACS change via the shelterwood method may solely focus on its final harvest which involved major overstory removal with a potential to reduce 30%-50% aboveground carbon storage (Conner et al., 2015; Scott et al., 2004).

1.4.4 Limitations and implications for future research

Results from this study provided additional support for achieving near-term carbon objectives in intermediate silvicultural treatments. However, this research focused solely on the carbon dynamics of live aboveground biomass. While most carbon market protocols only consider

aboveground carbon pools for inclusion in forest carbon projects (Urbano & Keeton 2017), comprehensive forest management planning should also account for carbon exchange from other sources to address climate change, which may include forest floor and belowground carbon pools, standing deadwood and coarse woody debris, emissions from logging activities and the lifetime fate of harvested timber (Fraser et al., 2023; Nunery & Keeton, 2009; Zhang et al., 2023; Zhao et al., 2022). Another limitation of this study was that the recent inventory period was only 13 years after the initial overstory harvest. Consequently, this research was unable to provide a full temporal assessment of FACS recovery from intensive management regimes, such as clearcutting, which may require a half a century of observation to assess recovery (Davis et al., 2009; Woods & Kern, 2022). As a 100-year project, future inventory data from HEE will be able to provide further insights into long-term forest carbon sequestration in hardwood forests in southern Indiana within the context climate variability.

Future research may also compare carbon stock changes among oak-hickory, maple-beech, and other forest types in response to different silvicultural treatments. One of the primary objectives of HEE is to develop silvicultural prescriptions to maintain and regenerate oak-hickory forests in southern Indiana (Kalb & Mycroft, 2013). With proper management, potential carbon stocks of oak-hickory forests in the CHR may reach their maximum potential carbon stocks at 345 Mg/ha, whereas maple-beech-birch forest types may reach their maximum at 212 Mg/ha (Birdsey et al., 2023a). For EA management, early attempts to regenerate oaks from clearcutting were more successful on xeric sites but unsatisfactory on mesic sites (Johnson et al., 2009; Swaim et al., 2016). Meanwhile, oak seedlings often faced challenges of competing with other early-successional species, such as tulip poplar and sassafras (Dey et al., 2008; Kellner & Swihart 2017). The shelterwood system, often accompanied by pre-harvest prescribed burns, was found to be more effective in promoting oak regeneration (Greenler & Saunders 2019). Compared with clearcutting that created a full-sunlight environment, shelterwood harvesting created intermediate light levels that favored the growth of oak species, which have moderate shade tolerance. In contrast, canopy gaps created from single-tree selection (UEA) were too small to favor regeneration of oak species. Insufficient light levels from this harvesting method instead promoted growth of shade-tolerant species such as maple. Comparisons of carbon stock change among different community types will provide further evidence of the effectiveness of oak regeneration from various management practices.

1.5 Conclusions

The average FACS in MMYSF were 115.3 ± 2.1 Mg/ha and had the potential to increase with less intensive silvicultural treatments from both EA and UEA management. Of all treatments tested in this study, single-tree selection (UEA) had no effect on FACS throughout all post-treatment inventories over 13 years and had the closest MAI to the untreated controls. For three-staged shelterwood treatment (EA), the loss of FACS from the first two cuts combined were offset by a decade of regrowth and reached its pre-harvest level by 2021, despite the shelterwood forests still had 20% less FACS than the untreated controls (NH) in 2021. The two prescribed burns (EA) in future clearcut and shelterwood sites also had no impact on FACS. However, intensive harvesting approaches such as clearcutting (EA) and patch cutting (UEA) resulted in the lowest FACS among all treatment types. Even after 13 years of recovery, treated forests only recovered to 26.9 ± 6.6 Mg/ha, which was approximately 30-37% of their pre-treatment levels. The untreated forest at MMYSF was a typical representation of middle-aged, secondary forests in the eastern US regrowing from historical land abandonment in the 1920s. A younger average stand age explained its rapid carbon accumulation compared to old-growth forests in the CHR. Overall, results from this study provided insights for understanding near-term FACS changes among different silvicultural treatments. Long-term inventory data from HEE will allow long-term evaluation of carbon stock dynamics in managed hardwood forests.

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CHAPTER 2. USING 3DEP LIDAR TO ESTIMATE FOREST ABOVEGROUND CARBON STOCKS

2.1 Abstract

Reliable quantification of forest aboveground carbon stocks (FACSS) is essential in climate-adaptive forest management. Although Light Detection and Ranging (LiDAR) technology has become an effective approach for FACS estimation, time-consuming and labor-intensive field data collection is still the dominant method used in Indiana. The overarching goal of this study was to investigate the feasibility of using the 3DEP LiDAR for multiscale FACS estimation in Indiana state forests. Random forest regression models were adopted to predict FACS across different spatial scales, silvicultural practices, and point densities. It was found that the overall model's root mean square error (RMSE) decreased from 25.43 Mg/ha (26%) to 16.74 Mg/ha (20%) when spatial scale expanded from plots to stands. This was attributed to insufficient plot-level positioning accuracy and magnified susceptibility to boundary effects for a plot than for a stand. It was also found that an average LiDAR point density at 2.0 points/m² demonstrated comparable performance (16.81 Mg/ha, 20%) on stand-scale FACS estimation as compared to a higher density of 7.7 points/m² (15.73 Mg/ha, 19%). This was likely because a stand was associated with a greater number of laser returns than a plot, thereby retaining the ability to capture the spatial variation in forests and preserving the overall prediction accuracy. However, the partitioned random forest model that solely included clearcut and patch cut sites generated 59% RMSE (17.82 Mg/ha), much higher than the result from another partitioned model with samples of all other treatments (14%, 13.76 Mg/ha), including control, single-tree selection, prescribed burning, and shelterwood. This was due to inaccurate LiDAR return classification outcomes from the 2011-2013 dataset, which created notable "vacuum" areas that led to biased canopy metric extraction. This phenomenon together with the averaging effect inherent from the random forest algorithm has collectively resulted in overestimated FACSS at both clearcut and patch cut sites. In conclusion, this study suggests the 3DEP LiDAR system is a prospective solution for multiscale FACS assessment in temperate hardwood forests in Indiana. The findings provide valuable insights for forest management, carbon monitoring, and decision-making processes in the context of climate change mitigation.

Keywords: Aboveground carbon stock, aboveground biomass, clearcut, co-registration error, disturbance, LiDAR, LiDAR return classification, point density, random forest, spatial scale

2.2 Introduction

2.2.1 Limitations of traditional approaches on AGB calculation

Reliable quantification of forest AGB is essential for the estimation of forest aboveground carbon stocks (FACSs). Forest AGB has traditionally been estimated with field-based approaches, such as destructive sampling and allometric models (Lu et al., 2016). Destructive sampling includes harvesting and measuring the dry weight of one or more sample trees. This approach gives the most accurate estimation results at the individual tree level, but it is impractical and introduces sampling errors when scaling up to a stand- or landscape- level. In contrast, allometric equations estimate biomass using easily measured tree attributes (e.g., DBH and height), which vary with taxonomic groups driven by wood specific gravity. This approach has been widely applied in the literature, resulting in numerous available databases of allometric equations (Chojnacky et al., 2014; Jenkins et al., 2003). Nonetheless, equations developed at one site may not work well on another. Environmental variations, such as soil, sunlight, topography, climate, past land-use, as well as stand management history, are all likely to induce different growth rates for the same species and add error to biomass estimations. Moreover, there is a notable lack of equations for many less-common species or non-native species, even within national-level databases (Jenkins et al., 2004). Furthermore, measurement of DBH and tree height still requires extensive field work, which is inherently time consuming and labor intensive. For example, the Forest Inventory and Analysis (FIA) program is tasked with providing plot-level information on a national scale to monitor ecological changes in U.S. forests. Although the FIA program receives \$66.8 million annually, only about 10-15% of forest plots in each state are re-measured either annually or on a five-year cycle (USDA FIA, 2016). In addition, external challenges such as harsh weather conditions, dense vegetation, in combination with human error, can further add difficulties for field data collection (Oh et al., 2022).

2.2.2 Comparison of LiDAR and other remote sensing tools in forest AGB estimation

Compared to traditional field survey methods, remote sensing technologies allow the measurement of trees across large spatial extents in a timely manner. Moreover, remote sensing data can not only provide summary statistics over a given area, but also generate a spatially explicit AGB “heatmap” for a landscape, thereby aiding in the identification of fine-scale changes in forest conditions. Remote sensors can broadly be categorized into two groups. The first group includes optical sensors capable of measuring multi-spectral, panchromatic, and hyperspectral radiation. These sensors do not directly emit electromagnetic radiation themselves, but instead record reflectance from objects. The extraction of spectral signals that capture the spatial patterns of plants (or the environment) allows the estimation of AGB. Extracted metrics include spectral bands at various wavelengths, vegetation indices, and transformed bands such as images created from principal component analysis and textural analysis (Chen, 2013; Lu et al., 2016). However, spectral signals tend to oversaturate, especially in closed-canopy forests with complex stand structures (Huang et al., 2021; Lu et al., 2012; Pflugmacher et al., 2014). In this case, additional data collection may not improve AGB estimation accuracy but rather increase computational complexity, leading to waste of resources and increased analysis time costs. Moreover, environmental fluctuations, including atmospheric conditions, soil moisture, vegetation vigor and phenology, and even shadows, can all influence the quality of the extracted metrics (Lu et al., 2016). In general, these passive sensors are better at capturing optical characteristics of vegetation and assessing horizontal canopy conditions, but they are incapable of measuring vertical canopy structure (Zolkos et al., 2013).

LiDAR is a type of active sensor that emits laser pulses towards objects and measures the time required for pulses to return. This capability enables LiDAR to achieve precise tree height measurements, which exhibit a robust correlation with forest AGB (Chandrasekaran et al., 2022). Traditionally, in-field height measurement relies on instruments such as hypsometers or clinometers; however, these methods may be prone to inaccuracies due to environmental factors such as steep topography and dense canopy cover. However, these shortcomings can be overcome by using LiDAR. Oh et al. (2022) revealed a strong correlation between LiDAR-estimated height estimates and field-measured heights. Moreover, laser pulses can penetrate forest canopies and enable the measurement of sub-canopy vegetation (Jarron et al., 2020). As a result, LiDAR-derived metrics were found to have a strong correlation with AGB for up to 1000 Mg/ha, which

far exceeded the typical saturation point of passive optical or radar sensors (Means et al., 1999; Zolkos et al., 2013). Furthermore, representation of vegetation vertical structure may be further improved by increasing point density. With point density exceeding 10 pts/m², individual tree crown segmentation becomes feasible, facilitating tree-level AGB estimation (Goldbergs et al., 2018). With point densities reaching 100 pts/m², the accuracy of individual tree segmentation can reach 98.6% (Qin et al., 2023). Owing to these advantages, LiDAR data have been favored for tracking forest structural changes in a landscape experiencing canopy disturbance (Boucher et al., 2020, Krisanski et al., 2021; Ma et al., 2017; Rex et al., 2020).

2.2.3 Factors that limit LiDAR-based FACS estimation accuracies

Although LiDAR technology offers remarkable efficiency and accuracy for AGB and FACS estimation, high point density LiDAR data acquisition over large areas can be very expensive. Forest owners and natural resource managers with constrained budgets must grapple with the decision of whether to procure low point density data covering a broad area or to invest in high point density data for a specific subset of smaller areas (Jakubowski et al., 2013). Fortunately, the USGS 3D Elevation Program (3DEP) offers a solution by providing comprehensive, nationwide high-resolution elevation data, encompassing both bare earth terrain and detailed 3D point clouds, free of charge. As of 2022, more than 80% of the United States had access to 3DEP LiDAR data or had data acquisition projects underway (Stoker & Miller, 2022). However, it is worth noting that the median point densities for most 3DEP data typically range from 3.6 to 4.2 pts/m². Despite this availability, to my knowledge, there have been no studies using 3DEP LiDAR data for FACS estimation. Thus, better understanding of point density impact on FACS estimation will provide crucial information to stakeholders navigating resource allocation dilemmas.

In addition, it should also be noted that a FACS predicting model developed for a specific forest may not exhibit high performance when applied to a different forest. This can be attributed to inherent variations in canopy structure, species compositions, and environmental conditions across forests. Prior research has necessitated the development of site-specific models based on forest cover type, age class, or disturbance regime to enhance FACS prediction accuracy (Anderson & Bolstad, 2013; Hernández-Stefanoni et al., 2020; Knapp et al., 2017; Zhao et al., 2016). However, many of these studies have focused on less disturbed forests, such as those

undergoing selective logging practices (Dalagnol et al., 2019; Pinagé et al., 2019; Rex et al., 2020; Swetnam et al., 2015). There is a lack of research that evaluates LiDAR's performance in disturbed forests, such as clearcut sites. Evaluating LiDAR's performance across different disturbance types can validate and improve the accuracy and applicability of FACS estimation methods.

Furthermore, sampling design can also affect LiDAR's performance in the prediction of FACS. A key sampling design parameter is the field (reference) plot size, which refers to the spatial extent used for aggregating or analyzing LiDAR data. This parameter is often interchangeably referred to as spatial scale (Frazer et al., 2010), which can be easily confused with spatial resolution pertaining to the ground area represented by each pixel in an image. Although numerous studies have discussed how spatial resolution influences FACS estimation (Deo et al., 2018; Guo et al., 2023; Johnson et al., 2022; Tao et al., 2016), less attention has been devoted to the influence of spatial scale. In general, smaller plots are better at capturing local variations in forest structure; however, the extracted LiDAR metrics are susceptible to the influence of extreme values and outliers (Hernández-Stefanoni et al., 2018; Næsset et al., 2015). Conversely, larger plots encompass greater spatial coverage and can capture a broader spectrum of population variability, but deploying larger plots involves increased fieldwork and associated expenses. In forest ecological studies, the determination of spatial scale is directly related to specific research objectives. For example, measuring the densities of overstory trees, rare species, or coarse woody debris typically necessitates larger sampling plots compared to assessing understory vegetation or basal area (Carrer et al., 2017). Moreover, even for quantifying the same attribute, the optimal spatial scale may vary across different vegetation types due to their distinct stand structures (Pinto et al., 2020). Thus, understanding how spatial scale impacts AGB and FACS estimation is important for refining research sampling design, thereby optimizing prediction accuracy and ground reference sampling efficiency.

Based on these limitations, this study aimed to investigate the reliability of using 3DEP LiDAR data for estimating FACS in Indiana's hardwood forests. To develop a comprehensive understanding of the 3DEP LiDAR's effectiveness, I used random forest regression models to evaluate the impacts of spatial scale, LiDAR point density, and different silvicultural practices on FACS prediction. My study objectives were as follows: 1) compare the overall FACS prediction accuracy between plot- and stand- scale analyses, 2) assess the localized FACS prediction accuracy

between clearcut forests and less disturbed forests, and 3) examine the FACS prediction accuracy between the 2011-2013 and the 2017-2020 3DEP LiDAR datasets, which have different point densities.

2.3 Methods

2.3.1 Study area

This study was focused on landscapes within Morgan-Monroe and Yellowwood State Forests (MMYSF) in Indiana (39.25° N, 86.39° W) (see **Figure 1-1** in Chapter 1). MMYSF have a temperate climate with a mean temperature of 24 °C (July) and -1 °C (January), as well as a mean annual precipitation of 1274 mm. This area remained unglaciated and is characterized by steep cliffs, rugged hills, and well-drained acid silt-loam soils (Homoya et al., 1985). Elevations across MMYSF ranges between 167 - 300 m, with an average slope of 18.5 degrees (Barnes et al., 2016; Shao et al., 2018). The species composition of MMYSF is largely oak (*Quercus* spp.) and hickory (*Carya* spp.), which collectively occupy 43% of the total property area and are reported to have occurred on this landscape for the past 7,000 years (Haulton, 2013; Wilkins et al., 1991; Jenkins, 2013). Typical upland overstory species include black oak (*Q. velutina*), white oak (*Q. alba*), northern red oak (*Q. rubra*), pignut hickory (*C. glabra*), and tulip poplar (*Liriodendron tulipifera*), whereas sugar maple (*Acer saccharum*) and American beech (*Fagus grandifolia*) dominate the midstory and understory (Lowney et al., 2015; Morrissey et al., 2015; Saunders & Arseneault, 2013; Spetich & Parker, 1998). MMYSF are typical representations of secondary forests within the Central Hardwood Region, recovering from forest degradation and fragmentation due to large-scale land clearing during the 1920s (Jenkins 2013; Urban, 2005). As of 2021, the average forest stand age across both forests was 75-110 years (Haulton, personal communication).

2.3.2 Field data collection and processing

Forest inventory data used in this study were collected as part of the Hardwood Ecosystem Experiment (HEE), which included nine forest management cores and their buffer zones across the MMYSF (**Figure 1-1**). Below is a brief overview of the experimental design, with detailed inventory protocols elaborated in section 1.3.2.

Each management core was approximately 90 hectares in size. A total of 519 and 705 0.1-ha sampling plots were systematically distributed across all management cores and were used for vegetation sampling during 2013-2014 and 2017-2018, respectively. Within each plot, all woody stems greater than 1.5 cm DBH (diameter of breast height) were recorded with attributes including species, DBH, crown class, azimuth and distance to the plot center, living condition, and stratum/light field. Tree height measurements were obtained for roughly 20% of total inventoried stems. Plot center coordinates were determined by a GPS device with an average deviation of 11 m.

There was a total of three even-aged cores (EA), three uneven-aged cores (UEA), and three control cores (NH) in total. Each EA/UEA core was further subdivided into multiple forest stands based on specific treatments received (**Table 2-1**). Each EA core included two clearcut stands, two shelterwood stands, four prescribed burning stands, as well as 12 untreated stands for future rotations. The average stand size for each EA stand was 4.5 ha. On the other hand, each UEA core featured one dominant single-tree selection stand and eight patch cut stands, with average stand sizes of 80.4 ha and 1.1 ha, respectively. No silvicultural treatment was implemented within the control cores. Since stand delineation was absent within the control cores, each core was considered a stand and merged with the untreated EA stands to form a new category termed "unmanaged". Most treatments were executed during 2008-2009, with exceptions such as shelterwood receiving a second cut during 2015-2016 and prescribed burning undergoing the first burn during 2015-2017 (**Table 2-1**). In 2006, topographic maps were used as a reference to create hand-digitized stand boundaries within ArcGIS pro. Subsequently, in 2013, stand boundaries of clearcut and patch cut sites were re-measured by a GPS device with an average positioning accuracy of 5 m (Saunders, personal communication).

Species-specific, DBH-based allometric equations (Chojnacky et al., 2013) were used to estimate individual live tree AGB and then scaled to plot-level AGB density, expressed in units of metric tons per hectare (Mg/ha). The plot-level AGB was then multiplied by 0.5 to obtain live forest aboveground carbon stocks (FACS; Hoover & Stout, 2007; Maclean et al., 2014; Anderson et al., 2023). Stand-scale FACS was calculated as the mean of plot-scale FACS within the stand. Chapter 1 demonstrated significantly low FACSs (14 - 39 Mg/ha) in clearcut and patch cut sites during 2013-2017, but there were minor differences between untreated controls and all other treatments during this period (89 - 109 Mg/ha) (More details were provided in **Table 1-3**).

Table 2-1. Summary of treatment stand characteristics, plot counts, and mean FACS between 2013 and 2017 at HEE sites.

Management	Stand type	Treatment descriptions	Avg. stand size (ha)	Total stands	Total plots (2013, 2017)	Avg # plots per stand	Mean FACS (Mg/ha)
Unmanaged (NH)	Control	No harvests in control units	86.1	3	118, 225	75	106 - 109
	EA unmanaged	EA sites that have not yet received any harvests	4.5	37	155, 155	4	97 - 100
Uneven-aged (UEA)	Single-tree selection	Target basal area: 16.1 - 22.9 m ² /ha	80.4	3	139, 201	67	99 - 102
	Patch cut	Complete overstory removal	1.1	24	17, 32	2	32 - 39
Even-aged (EA)	Clearcut	Complete overstory removal	3.9	6	22, 23	4	14 - 19
	Shelterwood	Target basal area: 13.8 m ² /ha (first cut) and 13.8 - 16.1/m ² (second cut)	4.7	6	24, 24	4	89 - 97
	Prescribed burning	Understory / midstory fires; stands for harvest in the future	4.8	12	44, 45	4	106 - 108

2.3.3 LiDAR data collection and variable extraction

The Indiana Statewide Quality-Level-2 (QL2) airborne LiDAR data were part of the USGS 3DEP project. LiDAR data across the entire state of Indiana were collected by a series of sensors (specifications were provided in **Table 2-2**), with a vegetated vertical accuracy of less than 0.3m (Stoker & Miller, 2022). The study area's 3DEP LiDAR data were acquired from the 2011-2013 and the 2017-2020 datasets (iDiF, 2020), with an average post spacing at 1.5 m and 0.7 m and a local point density of 2.0 pts/m² and 7.7 pts/m², respectively. Specifically, the Morgan-Monroe State Forest was surveyed during February-April 2011 and March 2018, while the Yellowwood State Forest was surveyed during March-April 2011 and April 2017. The 3DEP data products included raw laser point clouds (in LAS format), digital elevation model (DEM), and digital surface model (DSM). The LAS data files were calibrated and classified as ground and non-ground points. Hydro-flattened DEMs were generated using TerraScan, while DSMs were derived from the first return LiDAR pulse recordings of surface reflectance. Canopy height model (CHM) data were only available for the 2017-2020 dataset, so I derived CHM for the 2011-2013 dataset by subtracting DSM from DTM (Jiang et al., 2020), and subsequently trimmed off pixels with extreme values exceeding 50 m or falling below zero. Since the LiDAR data needed to be integrated with field data collected in 2013 and 2017, I simply refer LiDAR data collection years as '2013' and '2017' for the remainder of this thesis.

Due to differences in vegetation structure, data processing procedures and model selection criteria, there is no universal set of LiDAR metrics that can seamlessly apply to all studies (Chen, 2013). A general AGB estimation approach is to combine variables that capture both horizontal and vertical canopy variation (Fischer et al., 2019). These variables can be directly derived from point cloud data or LiDAR-derived products (e.g., CHM and DTM). In this study, a total of 38 commonly used height metrics were extracted as raster layers, including canopy height metrics (e.g., mean, median, maximum, standard deviation, and quadratic mean), canopy cover at seven height thresholds (2m, 5m, 10m, 15m, 20m, 25m, and 30m), aspect and slope, as well as height percentiles (5th, 10th, ..., 95th) and assorted summary statistics (**Table 2-3**; Duncanson et al., 2022; Knapp et al., 2017; Li et al., 2022; Rex et al., 2020; Shao et al., 2018). Canopy height metrics and canopy cover were created from the CHM. Slope and raw azimuth values were derived from the DTM. Since north and east aspects have greater productivity than other aspects, I used trigonometric-transformed aspect values to assign more weight on northeast aspect (Desta et al.,

2004, Beers et al., 1966). For height percentiles and other summary statistics, including mean, kurtosis, skewness, standard deviation, and median absolute deviation, these variables were extracted from the LAS files by using the "LAS height metrics" function in the 3D Analyst package within ArcGIS Pro. All 38 raster layers were resampled to a cell size of 3 m for both plot-level and stand-level variable extraction needs.

Table 2-2. Specifications of sensors used for 3DEP LiDAR data collection.

LiDAR system	Leica ALS50-II / ALS60, ALTM Gemini	Leica AL70 / ALS80, Optech Galaxy Prime, Riegl Q1560
Year	2011 - 2013	2017 - 2020
Max. number of returns	4	4 / infinite / 8 / 12
Minimum post spacing	1.5 m	0.6 - 0.9 m
Nominal point density	1.35 ppsm	2.3 - 2.7 ppsm
Flying height	1,981 m	1,846 - 2,377 m
Flight speed	130 knots	135 - 150 knots
Scan angle	40° (variable)	40 - 58 degrees
Scan rate	41.8 Hz (variable)	36 - 172 Hz
Pulse rate	150 - 200 kHz	250 - 600 kHz
Swath width	0 - 1.5 X altitude	1,442 - 1,930 m
Overlap percentage	25%	25% - 30%
Datum	NAD83 (horizontal); NAVD88 (vertical)	
Projection	Indiana State Plane East and West (FIPS 1301, 1302)	

Table 2-3. Descriptions of variables extracted from 3DEP LiDAR data.

Variable name	Description	Extraction source
cc2m, cc5m, ..., cc30m	Canopy cover (%) at 2m, 5m, 10m, 15m, 20m, 25m, 30m	CHM
CHM_mean	Mean top of canopy height	CHM
CHM_med	Median top of canopy height	CHM
CHM_max	Maximum top canopy height	CHM
CHM_sd	Standard deviation of top canopy height	CHM
QMH	Quadratic mean canopy height	CHM
slope	Slope in degrees	DTM
azi_trans	Transformed azimuth values (Beers et al., 1966)	DTM
ht_p5, ht_p10, ..., ht_p95	Height at 5th percentile, 10th percentile, ..., 95th percentile	LAS files
ht_mean	Height mean	LAS files
ht_stdev	Height standard deviation	LAS files
ht_skew	Height skewness	LAS files
ht_kurt	Height kurtosis	LAS files
ht_mad	Height median absolute deviation	LAS files

2.3.4 Development of random forest regression models

Random forest, a machine-learning algorithm, was used to model FACSs. A random forest model first bootstraps the original dataset and creates multiple subset samples. Each bootstrapped subset sample is then trained with predictor variables that are randomly selected without replacement to form a decision tree (**Figure 2-1**). The amount of predictor variables in each decision tree is usually far less than the total amount of variables. Finally, the output values from all decision trees are averaged as the final regression result. Unlike linear models, random forest models are not constrained by assumptions about the distribution of residuals, equal variance, or linearity and collinearity. Additionally, random forest models are particularly adept at handling large datasets with numerous input variables and are less susceptible to overfitting (Breiman, 2001). Multiple studies have demonstrated the effectiveness of random forest for forest AGB prediction. (Dai et al., 2021; Fassnacht et al., 2021; Jiang et al., 2020; Marchesan et al., 2020; Qin et al., 2023; Yu et al., 2022).

I developed two basic and four partitioned random forest regression models to evaluate the 3DEP LiDAR's performance on FACS estimation based on three factors: spatial scale, treatment types, and point density. Using samples from all treatment types and LiDAR data from both collection years of 2013 and 2017, two basic models were created to compare the prediction accuracy at the plot- and stand-scale. If the plot-level model outperforms the stand-level model, the remaining four partitioned models will operate at the plot-scale; conversely, if the stand-level model performs better, all remaining partitioned models will operate at the stand-scale. For the next two partitioned models, one model exclusively used clearcut and patch cut samples for model training, whereas the other one used samples from all other treatment types. Both models used samples from the two LiDAR datasets. For the final pair of partitioned models, one used samples exclusively from the 2013 dataset, while the other one used all samples from the 2017 dataset. This intricate framework was designed to scrutinize the nuanced interplay between spatial scale, treatment types, and LiDAR data characteristics in FACS estimation.

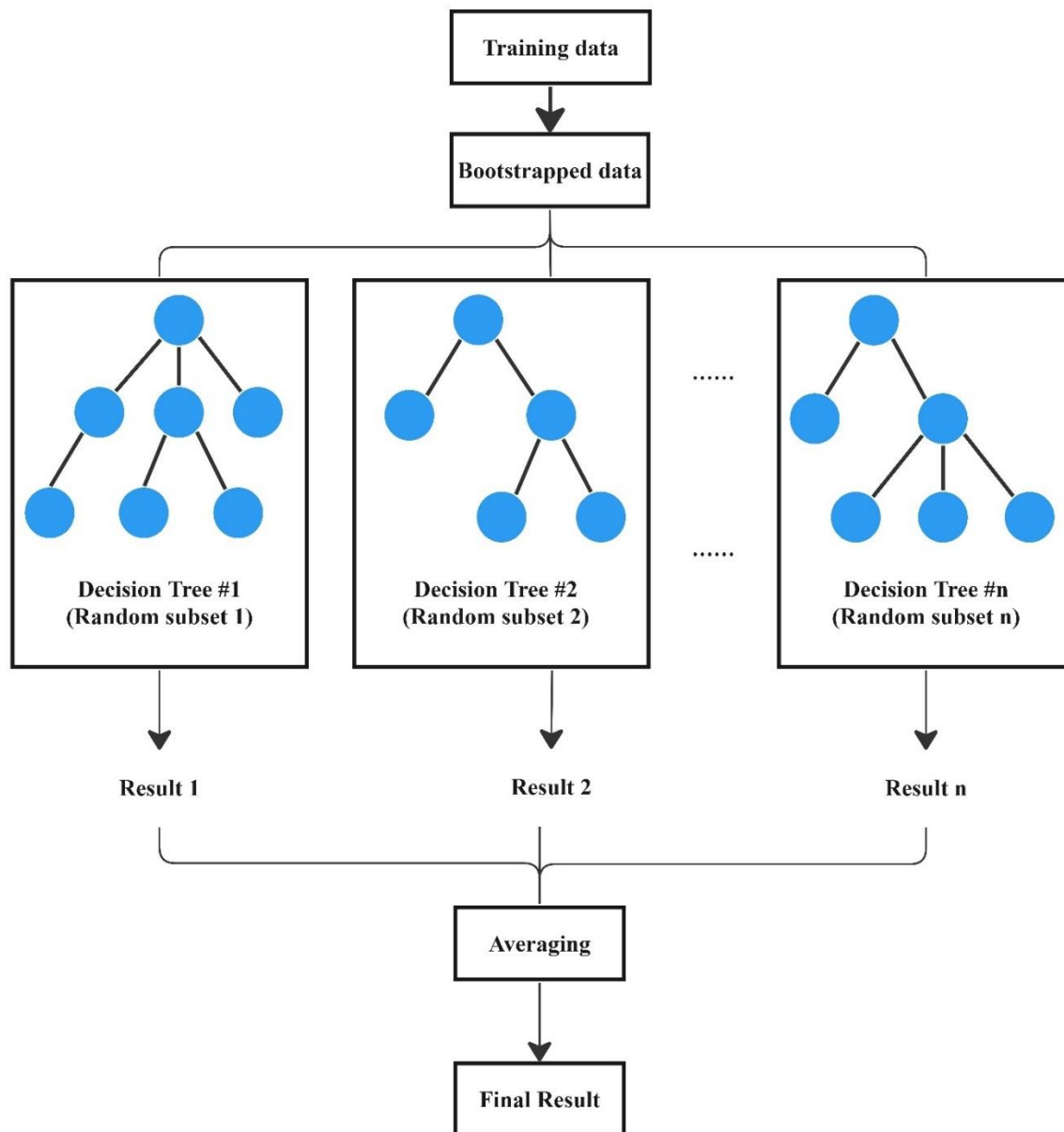


Figure 2-1. Simplified diagram of random forest regression modeling.

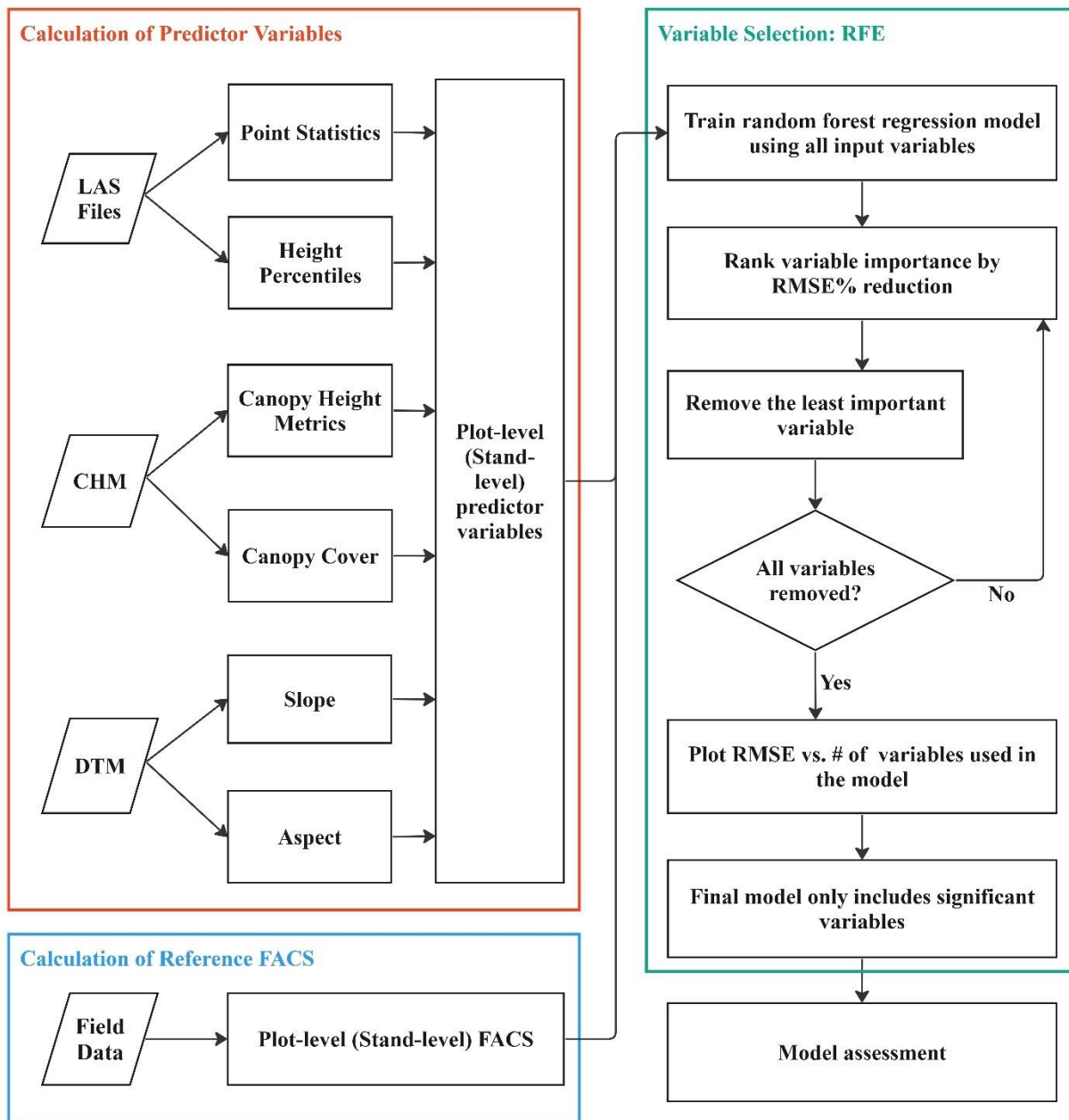


Figure 2-2. Diagram of creating a plot-level (or stand-level) random forest model and variable selection.

2.3.5 Variable important analysis and selection

To enhance model interpretability and reduce noise, it is essential to employ an effective variable selection technique to choose the most important LiDAR variables for FACS prediction. In this study, Recursive Feature Elimination (RFE) was adopted as the variable selection algorithm for the random forest regression models. RFE is a powerful method for optimizing model performance by identifying important variables, eliminating variables that introduce noise, and mitigating the impact of correlated variables on importance ranking (Demarchi et al., 2020; Gregorutti et al., 2017; Pullanagari et al., 2018). At the beginning, the original data were partitioned into a training set (80%) and a testing set (20%). All predictor variables were used to create the initial random forest model with the training set. Importance values were then calculated and ranked for all predictor variables, based on the criterion of reducing root mean square error (RMSE) in regression. The variable with the lowest importance score was subsequently removed from the model. The remaining variables were then retrained and re-ranked in the pruned model (**Figure 2-2**). This iterative procedure continued until a predefined number of variables (or none) were left for the final model.

To ensure the robustness of RFE, a 10-fold cross-validation was used as the outer resampling method and was repeated five times. This was implemented by using the "rfeControl" function in the R "caret" package. The outcomes of RFE were visualized by plotting the average RMSE over subset size that represented the number of variables used in the model. In this study, the optimal number of variables was determined based on two conditions: if the RFE curve exhibited a clear inflection point, all variables up to and including the inflection point would remain as final variables for random forest model training; if not, all variables up to the point that achieved the lowest RMSE would be retained. This approach aimed to balance model simplicity and RMSE reduction.

2.3.6 Model assessment and FACS mapping

After determining final variables to use for each model, three widely used error measurements were computed for the training set, including R^2 , RMSE, and relative RMSE (%) (Jiao et al., 2023; Li et al., 2020; Mahoney et al., 2022; Mauya et al., 2015). The RMSE% was calculated as the ratio of RMSE to mean observed FACS value. A higher R^2 value and lower

RMSE and RMSE% indicate a better model performance. To visualize the predicting accuracy of FACS at full data range, scatter plots of predicted versus reference FACS values were created for each model. Data points closely aligned with the identity line (the 1:1 line where predicted values equal the reference values) indicate accurate predictions, whereas the dispersion of data points around this line indicates a high variance. The above procedures were repeated in the testing set of each model. If the model's performance mirrors that of the training set, it shows confidence in the model's ability to make accurate predictions on unseen data. Conversely, disparities between the training set and testing set suggest poor generalization to unused data, potentially indicative of overfitting. By comparing the performances between basic plot and stand scale models, the better model was selected to generate FACS prediction maps of the study area for 2013 and 2017.

2.4 Results

2.4.1 Final variables for regression modeling

The RFE analysis revealed a distinctive inflection point for all six models, which was represented as a black dot (**Figure 2-3**). Consequently, all variables up to this inflection point were retained for final regression modeling, while those beyond it were deemed insignificant and removed. A drastic decrease of RMSE was observed in the *Plot_basic* model (light red) by using only three variables: CHM_max, cc20m, and CHM_sd (**Table 2-4**). The *Stand_basic* model (dark yellow) included seven variables: cc20m, CHM_max, cc30m, CHM_sd, cc15m, QMH, and ht_skew. As *Stand_basic* model had consistently lower RMSEs than the *Plot_basic* model, all partitioned models were configured at the stand-scale. Despite slight differences in variable importance rankings, the *Stand_2013* (green), *Stand_2017* (cyan), and *Stand_others* (pink) models had RFE curves highly similar to the *Stand_basic* model, with the *Stand_others* model having the lowest RMSE among all models (**Figure 2-3**). However, for the *Stand_clearcut* model, an unusual increase in RMSE was observed when more than two variables were used, indicating that only topographic variables (azi_trans and slope) were deemed considered significant in this model.

Table 2-4. Final variables for basic and partitioned regression models.

Model name	# variables	Final variables
<i>Plot_basic</i>	3	CHM_max, cc20m, CHM_sd
<i>Stand_basic</i>	7	cc20m, CHM_max, cc30m, CHM_sd, cc15m, QMH, ht_skew
<i>Stand_2013</i>	5	cc20m, CHM_sd, CHM_max, cc15m, cc25m
<i>Stand_2017</i>	5	Ht_kurt, cc25m, ht_skew, cc20m, ht_p90
<i>Stand_clearcut</i>	2	azi_trans, slope
<i>Stand_others</i>	4	ht_skew, CHM_sd, cc10m, CHM_max

Note: Variables were listed by the ranking of RMSE reduction.

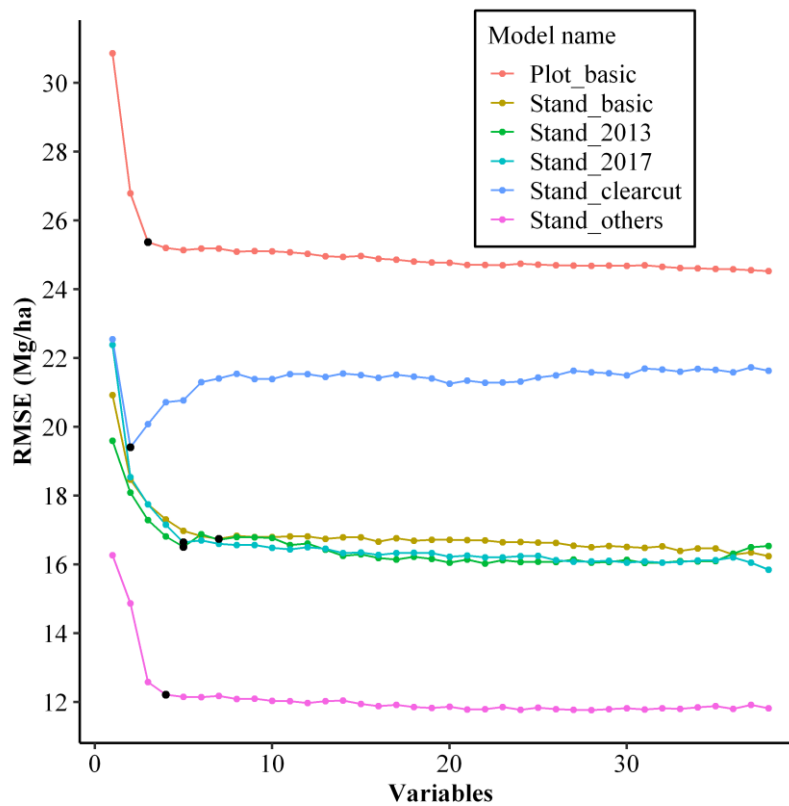


Figure 2-3. Results of recursive feature elimination (RFE) on basic and partitioned random forest regression models. A basic model used all samples in both LiDAR years, whereas a partitioned model only used partial samples or a single LiDAR year. The inflection point was labeled with a black dot.

2.4.2 Comparisons of basic and partitioned FACS estimation models

Even though the *Plot_basic* model was trained on 944, while the *Stand_basic* model used only 133 samples, the latter model exhibited superior predicting accuracy (**Table 2-5**). Specifically, the *Stand_basic* model explained 79% of the FACS variations and a 16.76 Mg/ha (20%) RMSE, whereas the *Plot_basic* model explained only 50% of the FACS variations with a higher RMSE of 25.43 Mg/ha (26%). Both models demonstrated a robust correlation between predicted and reference FACS values, albeit with data points more densely clustered around the 1:1 line for the *Stand_basic* model (**Figure 2-4**). However, the performance of the basic models appeared to vary across different stand types. A clustering effect was observed in the *Stand_basic* model, with the patch cuts and clearcuts cluster displaying higher variance compared to the cluster comprising all other treatments (**Figure 2-4; Figure 2-5**). Moreover, a notable overestimation of FACS was found for clearcut and patch cut samples with reference FACS values < 20 Mg/ha. Remarkably, both basic models demonstrated consistent performance between their respective training and testing sets, indicating the robustness to overfitting (**Table 2-5, Figure 2-4**).

The partitioned models yielded varying results. By retaining only topographic variables, the *Stand_clearcut* model achieved an RMSE of 17.82 Mg/ha, which was slightly higher than the *Stand_others* model (RMSE = 13.76 Mg/ha). However, the relative RMSE for the *Stand_clearcut* model (59%) was substantially higher than the *Stand_others* model (14%), and even exceeded that of the *Plot_basic* model (26%). Despite this, data points of *Stand_clearcut* model appeared to be more closely distributed along the 1:1 diagonal line than those of the *Stand_basic* model (**Figure 2-6**). However, the overestimation issue persisted, particularly for clearcuts and patch cuts. In contrast, data point distribution for the *Stand_others* model largely resembled that of the *Stand_basic* model, albeit with a slightly improved fitness. Both treatment-specific models demonstrated similar performance between their training and testing sets.

Interestingly, the overall model performance was not significantly impacted by LiDAR point density. Both the *Stand_2013* and *Stand_2017* models exhibited comparable RMSE values of 16.81 Mg/ha (20%) and 15.73 Mg/ha (19%), respectively. Furthermore, the distribution patterns of predicted values relative to reference values were quite similar for both models (**Figure 2-7**). Consistent with the trends observed in the other models, the performance of the testing set closely mirrored that of the corresponding training set for both the *Stand_2013* and *Stand_2017* models.

Table 2-5. Evaluation of basic and partitioned random forest models for FACS estimation.

Model name	Training set (80%)				Testing set (20%)			
	N	R ²	RMSE (Mg/ha)	RMSE (%)	N	R ²	RMSE (Mg/ha)	RMSE (%)
<i>Plot_basic</i>	944	0.50	25.43	26	236	0.48	27.39	28
<i>Stand_basic</i>	133	0.79	16.74	20	32	0.78	16.78	20
<i>Stand_2013</i>	64	0.77	16.81	20	15	0.75	16.79	19
<i>Stand_2017</i>	70	0.83	15.73	19	16	0.84	14.57	18
<i>Stand_clearcut</i>	36	0.40	17.82	59	8	0.55	15.77	54
<i>Stand_others</i>	97	0.25	13.76	14	24	0.38	13.49	13

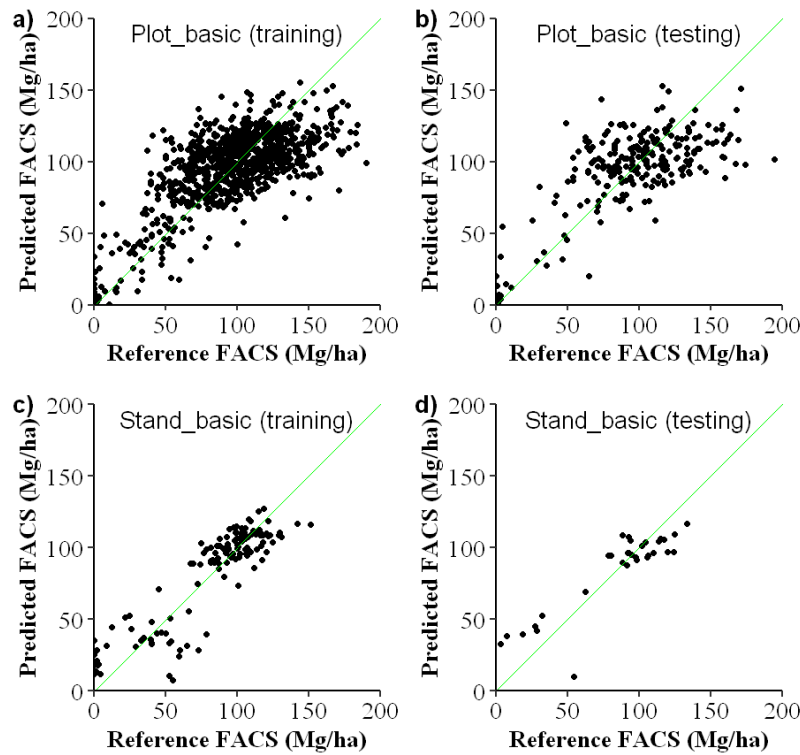


Figure 2-4. Scatter plot of predicted vs reference FACS values of the *Plot_basic* (top) and the *Stand_basic* model (bottom). The solid diagonal line represents a 1:1 relationship between variables.

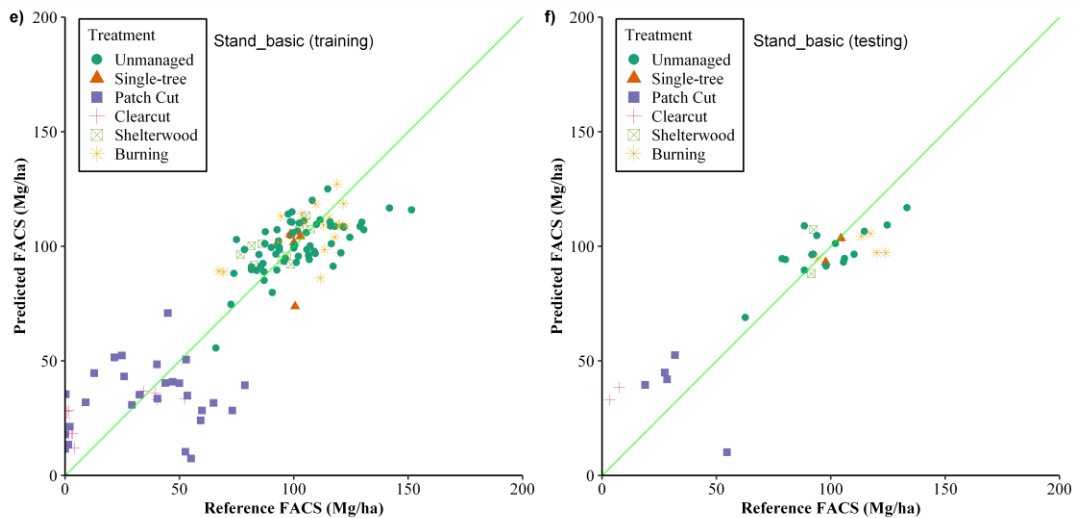


Figure 2-5. Scatter plot of predicted vs reference FACS values of the *Stand_basic* model, labeled by treatments. The solid diagonal line represents a 1:1 relationship between variables.

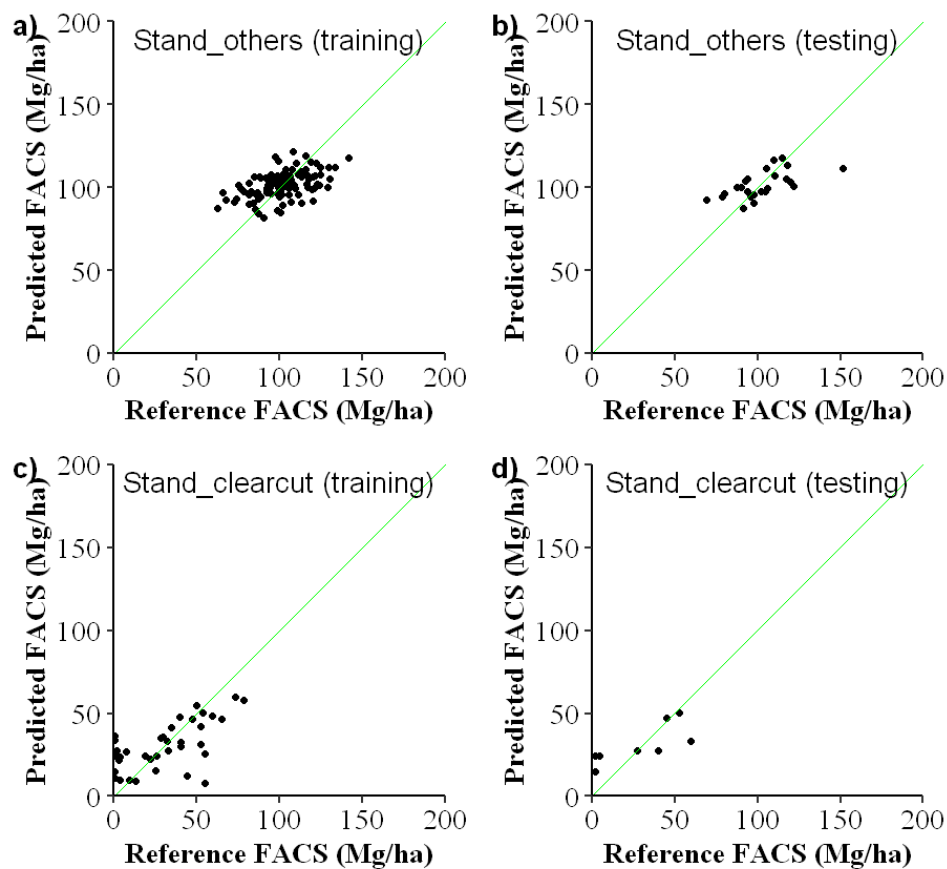


Figure 2-6. Scatter plot of predicted vs reference FACS values of the *Stand_others* (top) and the *Stand_clearcut* model (bottom). The solid diagonal line represents a 1:1 relationship between variables.

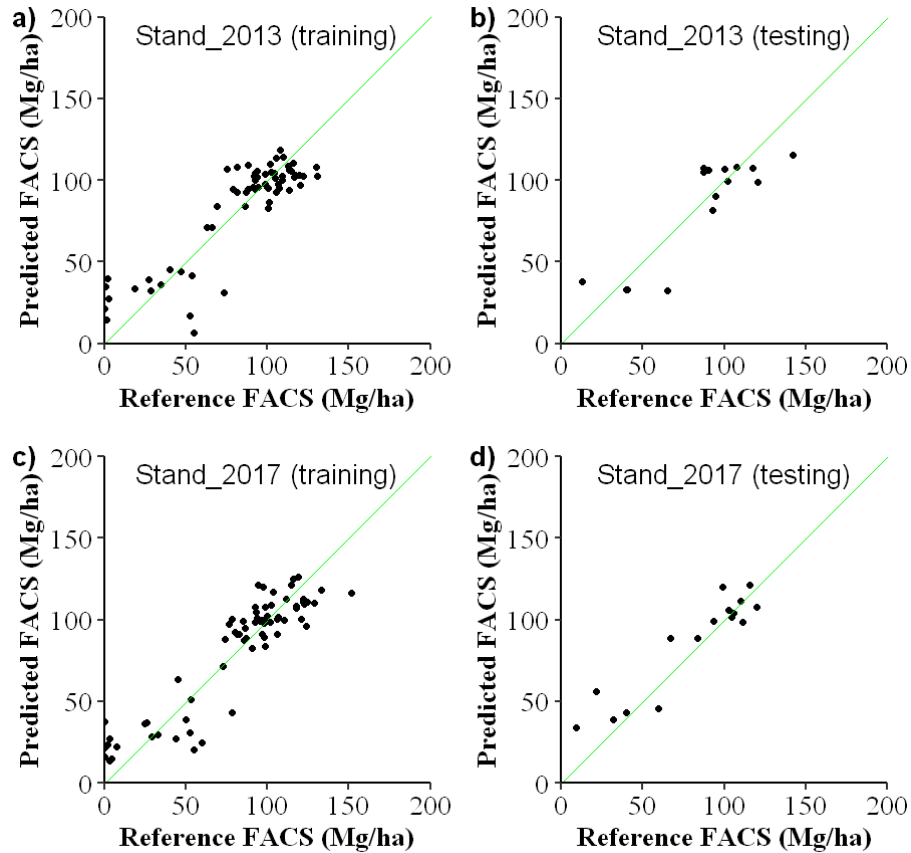


Figure 2-7. Scatter plot of predicted vs reference FACS values of the *Stand_2013* (top) and the *Stand_2017* model (bottom). The solid diagonal line represents a 1:1 relationship between variables.

2.4.3 Observations on FACS prediction maps

The *Stand_basic* model was selected to create FACS prediction maps of the entire HEE sites for both 2013 and 2017. The original FACS map in 2013 included a substantial amount of “No data” areas and was particularly noticeable in unit 5-9 (**Figure 2-8**). While units 1-4 had fewer blank regions, the FACS distributions appeared highly irregular, especially in unit 1 and much of unit 2. The FACS distributions of Unit 3 and 4 were relatively less impacted. I attempted to remediate this issue by replacing the blank regions with new pixel values by considering their neighbor pixels, but the “striping patterns” persisted across many study areas (**Figure 2-9**). In contrast, the FACS prediction map for 2017 did not exhibit such “striping” issues or noisy FACS distribution patterns (**Figure 2-10**). Regrowing forests resulting from clearcut harvests in even-aged units (3, 6, 9) and patch cut harvests in uneven-aged units (1, 7, 8) were clearly displayed

with low FACS values (red) as compared to adjacent treatment sites such as untreated controls, which displayed high FACS values (dark green).

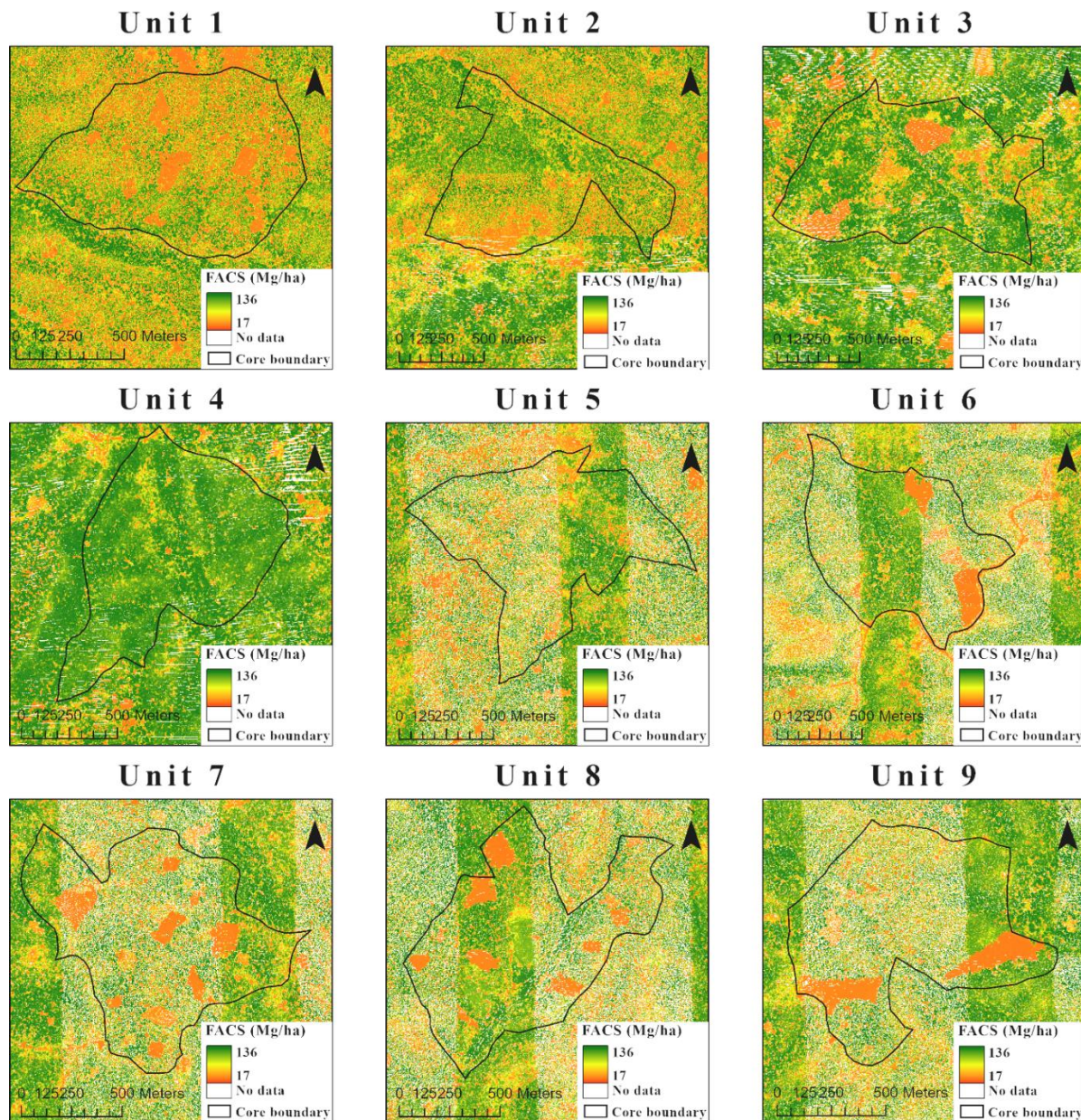


Figure 2-8. Raw maps of predicted forest aboveground carbon stocks (FACS) of nine research cores at HEE in 2013. Unit 1, 7, and 8 received uneven-aged management; Unit 3, 6, and 9 received even-aged management; Unit 2, 4, and 5 served as untreated controls.

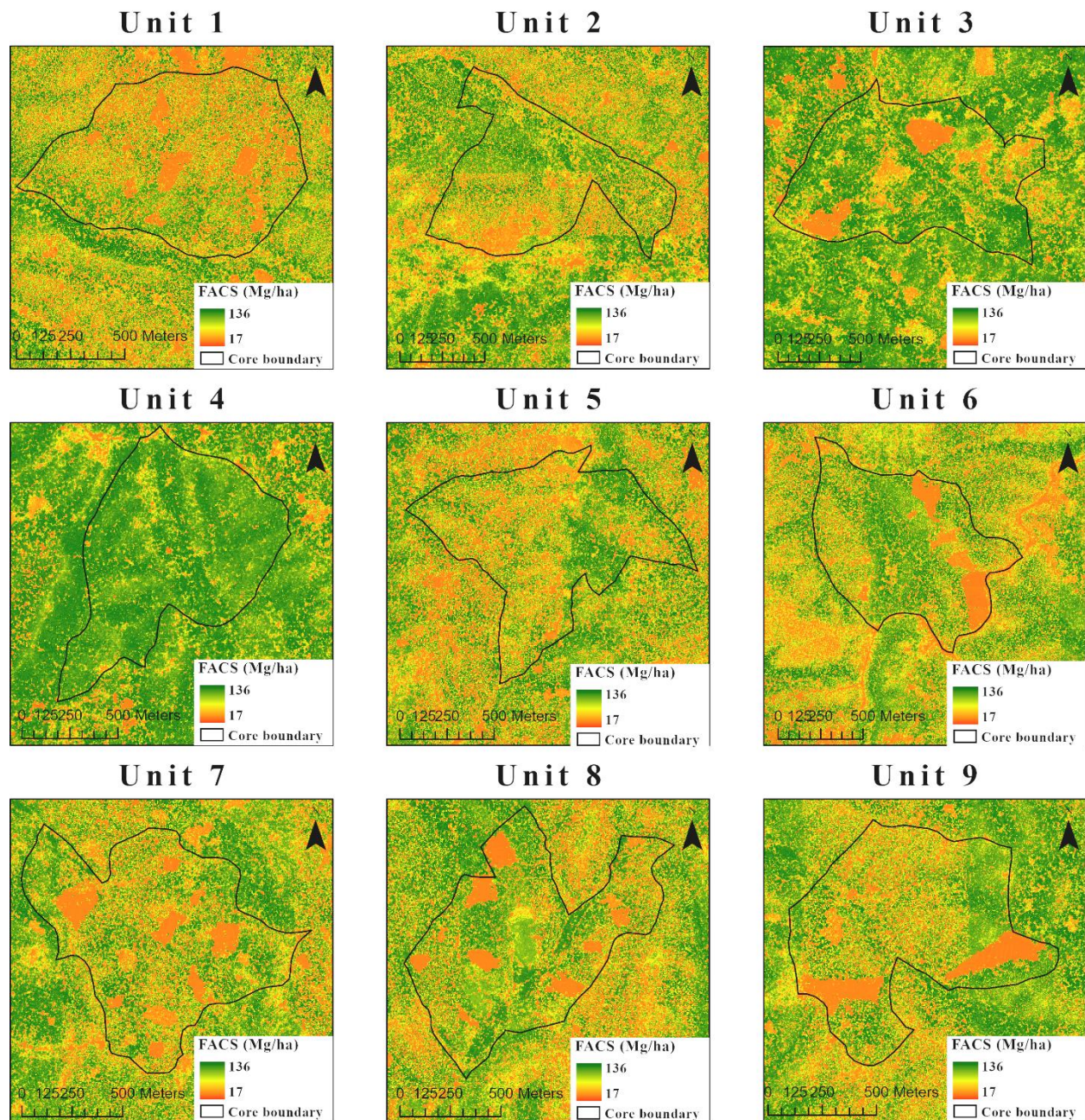


Figure 2-9. “Nibbled” maps of predicted forest aboveground carbon stocks (FACS) of nine research cores at HEE in 2013. The “No data” gaps were filled via “Nibble” function in ArcGIS pro. Unit 1, 7, and 8 received uneven-aged management; Unit 3, 6, and 9 received even-aged management; Unit 2, 4, and 5 served as untreated controls.

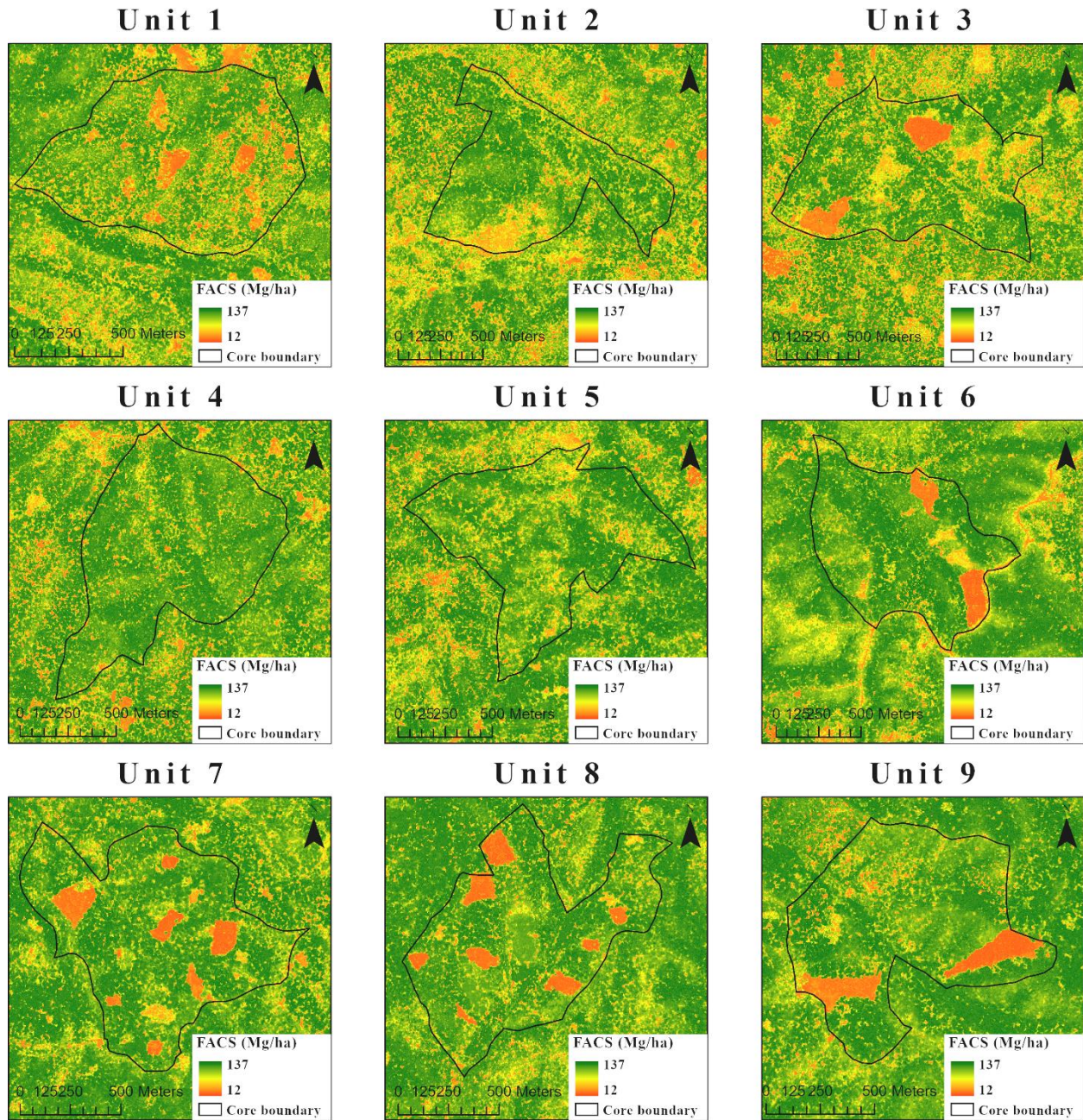


Figure 2-10. Maps of predicted forest aboveground carbon stocks (FACS) of nine research cores at HEE in 2017. Unit 1, 7, and 8 received uneven-aged management; Unit 3, 6, and 9 received even-aged management; Unit 2, 4, and 5 served as untreated controls.

2.5 Discussion

2.5.1 Effect of spatial scale on FACS estimation

There are two main reasons why the stand-scale model outperformed the plot-scale model. Firstly, stand-level analyses were less susceptible to edge-induced error than analyses at the plot level. The "edge effect" here referred to the influence of trees near the sample plot (or stand) border. A tree growing outside the border would not be tallied in field measurements, but part of its crown might be captured by LiDAR if it extended into the sample area. Conversely, a tree within a plot would be included in the field tally, but any part of its crown outside the plot boundary would not be included in the LiDAR data analysis. Such mismatches likely introduced noise into FACS predictions and tended to be more prominent at the plot level than the stand level. This discrepancy would diminish as the plot size increases. Knapp et al. (2021) reported that up to 40% AGB estimation uncertainty was contributed by edge-induced error at a 0.04 ha scale, but it diminished at a 1 ha scale. Given that the smallest stands in this study were patch cut stands with an average size of 1.1 hectares (**Table 2-1**), the impact of edge-induced errors in stand-scale models should be minimal.

Another substantial source of uncertainty was co-registration error, which was often concurrent with edge-induced errors (Gobakken & Næsset, 2009). Co-registration error referred to the spatial misalignment between ground samples and their corresponding LiDAR samples. The average geo-positioning accuracy of plots was 11.4 m, almost two-thirds of the plot radius (17.83 m), suggesting plot-level LiDAR variables may not represent the true plot conditions. This issue could be problematic for plots located between (or near) different treatment sites, where extracted LiDAR metrics for a patch cut plot, for example, might inadvertently represent closed-canopy structure from trees under single-tree selection treatment. In contrast, LiDAR variables extracted for stand-level models were less likely to reflect canopy structure outside harvest boundary because the average positioning accuracy of stand boundaries was 5 m. Gobakken & Næsset (2009) found that for a 5 m position error, increasing plot size from 0.02 ha to 0.04 ha would improve the overlap of the ground-true plot area from 60.8% to 72.0%, respectively. As the minimum average stand size for HEE sites was 1.1 hectares, the co-registration error should be negligible for evaluating whole-stand conditions.

Overall, the results of this study supported the argument that increasing spatial scale could improve LiDAR-based FACS estimation accuracy. Early publications reported an exponential decay relationship between plot size and model performance (Zolkos et al., 2013). Frazer et al. (2011) produced an RMSE% reduction from 47% to 34% by increasing plot size from 0.03 ha to 0.2 ha. Hernández-Stefanoni et al. (2018) observed an increase of R^2 (0.38 to 0.75) by increasing plot size from 0.01 ha to 0.1 ha. Mauya et al. (2015) demonstrated that increasing plot size from 0.02 ha to 0.3 ha accompanied by a substantial decrease in RMSE% from 64% to 29%. Knapp et al. (2017) even reported a drastic RMSE% reduction from $> 30\%$ to $< 10\%$ with spatial scale increased from 0.04 ha to 4 ha. In this study, switching spatial scale from plots to stands brought a decrease of RMSE% by 6-8% as well as an increase of R^2 by 30%. This indicates the potential of using stand-scale models for future AGB and FACS estimations.

It's important to note two significant differences between this study and others that evaluated the impact of plot size on AGB estimation. Firstly, stand size and shape were not controlled in this study. While most stands were of a similar size, a few stands were exceptionally large due to the lack of stand delineation in uneven-aged and control cores (**Table 2-1**). Additionally, as stand boundaries often followed ridges and valleys (Kalb & Mycroft, 2013), each stand often has a complex geometric shape as opposed to the regular shapes typically used for sampling, such as circles or rectangles. Nonetheless, while the boundary effect could be minimized by using circular plots, which have the smallest perimeter-to-area ratio among all other shapes, the choice of sample plot shapes became less important when the field plot area was large enough. Secondly, there was a difference in field sampling effort. Previous studies that compared the impact of spatial scale on AGB sampled all trees within the sample plot regardless of plot size. However, in this study, the reference FACS was calculated as the average of plot-level FACS within the stand, with most stands including four plots per stand (**Table 2-1**). Compared with counting all trees within the stand, aggregating plot-level estimates into larger units may still reduce spatial scale impact without increasing additional field work for large-scale ground sampling.

2.5.2 Inadequate performance predicting FACS of clearcuts and patch cuts

It was surprising that only azimuth and slope variables could mitigate error for the *Stand_clearcut* model. This phenomenon likely stemmed from inaccurate LiDAR point

classification at clearcut and patch cut sites. In this study, the "LAS height metrics" function in ArcGIS Pro was used for height metrics extraction, a process that relied on non-ground returns. However, in the 2011-2013 raw LiDAR point cloud, a significant portion of laser points within clearcut and patch cut sites was classified as ground returns, leaving minimal non-ground returns for canopy height metric extraction and model training (**Figure 2-11**). In other words, considerable areas within the clearcuts and patch cuts lacked sufficient height pixels for model training. When computing height statistics within a polygon, zonal analysis only considered pixels with actual height values and discarded the "No Data" areas. The very few available pixels primarily originated from residual trees in the middle of the harvest opening or trees near harvest boundaries, resulting in biased extraction outcomes. In contrast, DTM-derived aspect and slope, which were derived from ground returns, were not influenced by the classification issue and hence remained the only two variables suitable for model training. Interestingly, this classification issue was less pronounced for the 2017 LiDAR data, at least within the HEE sites (**Figure 2-12**). I proposed that the absence of non-ground returns at the clearcuts and patch cuts was because the regrowth forests following the 2008 harvest were too short to be distinguished from ground elevations by the 2011 LiDAR survey, but they were successfully detected by the 2017-2018 LiDAR survey as trees were taller. Unfortunately, the point classification method was not specified in the Indiana statewide LiDAR report, so it is unclear what height threshold was used to separate non-ground returns from ground returns. In practice, there has been no consensus on height threshold selection, but it typically ranges between 0 to 3 m (Anderson & Bolstad, 2013; Chan et al., 2021; García et al., 2010; Huang et al., 2019; Ma et al., 2018). Few studies have evaluated how the height threshold selection may influence AGB estimation, although some reporting the inclusion of ground returns could provide more information about the lower canopy and improve AGB predicting accuracy on heterogeneous forest landscapes (Chan et al., 2021; Mahoney et al., 2022). Given that nearly one third of all returns in 3DEP LiDAR data are classified as ground returns (Stoker & Miller, 2022), future research may further investigate whether information derived from ground returns can contribute to forest regrowth monitoring.

In contrast, areas outside the clearcuts and patch cuts were generally less impacted by the classification issue (**Figure 2-11**), which may explain why the *Stand_others* model that included predominantly overstory stands had a much lower error (**Table 2-5**). This was also supported by the model's final variable selection, in which all variables remained were canopy height metrics

rather than topographic variables (**Table 2-4**). However, removing clearcut and patch cut stands from model training also resulted in a low R^2 . This implied that the model did not capture the full extent of data variability, which likely includes species composition differences among treatment stands. Although laser returns are informative in height metrics extraction, they are incapable of capturing variations between different tree species. Future research may integrate LiDAR technology with optical sensors to infuse species-specific information into forest AGB and carbon stock modeling (Dalagnol et al., 2019; Jiao et al., 2023; Liu et al., 2019; Urbazaev et al., 2018).

Another important factor that contributed to a high RMSE% for the *Stand_clearcut* model was the overestimation of FACS for stands with less than 10 Mg/ha reference FACS values, which accounted for approximately 25% of total training samples (**Figure 2-6 c**). The random forest algorithm inherently tends to overestimate extremely low values and underestimate exceedingly high values (Breiman, 2001; Liu et al., 2019; Zhang & Lu, 2012). Unlike a linear model, a random forest cannot extrapolate. The final prediction generated by a random forest regression model is an average value of all decision trees, which reduces data variation but contracts the prediction range and results in overestimation at the lower end and underestimation at the upper end. This overestimation pattern was also observed in *Stand_basic*, *Stand_2013*, and *Stand_2017* models that included samples of all treatment types (**Figure 2-5**, **Figure 2-7**), but it did not significantly impact overall model performance because these overestimated samples only accounted for less than 10% of total sample size and represented a minor proportion of the total FACS range (e.g., 0-150 Mg/ha) covered by the model. As the model predictions were generally accurate for samples with > 80 Mg/ha FACS which dominate the FACS range, the influence of overestimation in the lower range was greatly lessened. Despite its limitations, random forest models remain a favored choice for AGB prediction. Given a sufficient sample size, the combined application of LiDAR and the random forest model may yield the lowest prediction errors in comparison to other machine learning algorithms and linear regression models (Dai et al., 2021; Fassnacht et al., 2014; Qin et al., 2023). In this study, all stand-scale models that included a full range of treatment samples successfully constrained the RMSE% within the range of 15 - 20%. This achievement aligns with findings from other LiDAR-based AGB studies using the random forest algorithm, where RMSE% typically falls within the range of 20 - 40% (Dai et al., 2021; Jiang et al., 2020; Ma et al., 2023; Marchesan et al., 2019; Torre-Tojal et al., 2022). This congruence supports the effectiveness and reliability of the stand-scale models in landscape-level FACS estimation.



Figure 2-11. Binary maps of presence (light green) and absence (white) of the 2013 3DEP LiDAR non-ground returns at each research core at HEE. Dashed polygons indicate areas of clearcut and patch cut harvest. Solid polygons indicate research core boundaries.

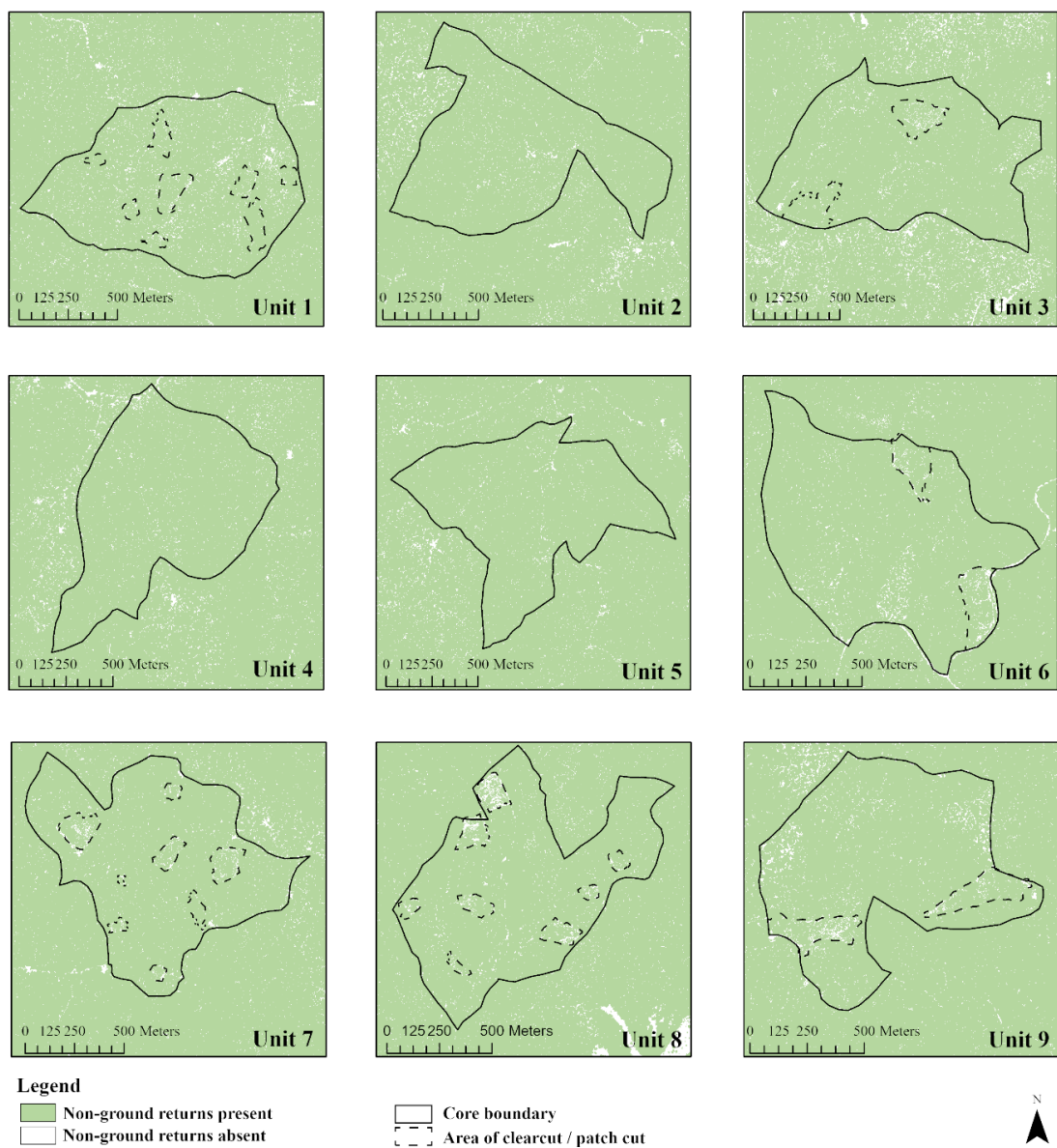


Figure 2-12. Binary maps of presence (light green) and absence (white) of the 2017 3DEP LiDAR non-ground returns at each research core at HEE. Dashed polygons indicate areas of clearcut and patch cut harvest. Solid polygons indicate research core boundaries.

2.5.3 Effect of LiDAR point density on FACS estimation

Highly similar FACS predicting accuracies between the *Stand_2013* model and the *Stand_2017* model supported the viability of using reduced-density LiDAR data for community-level FACS estimation. Previous research indicated that, at a 0.05 ha scale, increasing point density beyond 1 pt/m² would not significantly improve the predictive accuracy of canopy cover metrics (Guo et al., 2023; Jakubowski et al., 2013; Ma et al., 2017). Given that the average point densities used in this study were 2.0 pts/m² and 7.7 pts/m² for *Stand_2013* and *Stand_2017*, respectively, these densities have surpassed the threshold. Therefore, it is reasonable to infer that the structural characteristics of the canopy were adequately preserved under both scenarios.

Furthermore, even if the 2.0 pts/m² density provided less detailed information regarding canopy structure per pixel, any potential impact on canopy height metrics extraction should have been mitigated by increasing the spatial scale. Huang et al. (2013) demonstrated that irrespective of point density, models at a spatial scale of 1.0 ha consistently exhibited approximately 5% lower RMSE compared to models at a 0.25-hectare scale for AGB estimation. This was because larger spatial scales entail a greater number of returns per sample, thereby enhancing their capability to capture spatial variation within structurally complex forests (Chan et al., 2021; Hudak et al., 2012). Given that all forest stands considered in this study exceeded 1 ha in size, any discrepancy in point density should have negligible influence on stand-scale FACS estimation.

While the overall model performance between the two LiDAR datasets remained highly similar, caution should be taken when using the 2013 FACS prediction map to capture spatial variations at the local scale. Inconsistencies in point density were prevalent throughout the study area, resulting in significant "NoData" regions (**Figure 2-8**). Oh et al. (2022) identified several factors contributing to the presence of low LiDAR density strips, including improper LiDAR sensor configuration, low flight altitude, unexpected weather conditions, or abrupt aircraft maneuvers. Conversely, the few areas with higher LiDAR density accounted for only a small proportion of the study area and were likely a consequence of overlapping LiDAR scans. While these blank regions may potentially be filled by considering neighboring cell values, this approach could introduce noisy pixels and complicate visual interpretation (**Figure 2-9**). Decreasing map resolutions can create a smoothing effect to mitigate the impact of noisy pixels. However, further investigation is needed to understand the tradeoff between resolution reduction and loss of details. Nonetheless, it is beyond the scope of this study and may be explored in future research endeavors.

In contrast, the 2017 FACS prediction map (**Figure 2-10**) exhibited superior interpretability owing to its higher point density and reduced flightline positioning errors (Jung & Jung, 2023; Scott et al., 2022). Given the lack of significant differences in FACS between 2013 and 2017, I recommend using the 2017 map as an alternative to demonstrate overarching trends in FACS distributions across HEE sites.

2.6 Conclusions

This study used random forest models to investigate the feasibility of using 3DEP LiDAR data for predicting FACSs across variations in spatial scale, silvicultural treatments, and LiDAR point density. It was found that the increase in spatial scale from plots to stands greatly reduced RMSE from 25.43 Mg/ha (26%) to 16.74 Mg/ha (20%). This reduction was attributed to the mitigated effects of co-registration error and the boundary effect at the stand level compared to the plot level. Forest landowners and resource managers facing financial constraints in acquiring high-precision GPS devices may alleviate this limitation by transitioning from a plot-level to a stand-level spatial scale to optimize FACS prediction accuracy. Additionally, it was observed that an average LiDAR point density of 2.0 pts/m² yielded comparable prediction accuracy (16.81 Mg/ha, 20%) to that of 7.7 pts/m² (15.73 Mg/ha, 19%) across the entire study area. This phenomenon may also be associated with spatial scale dynamics, wherein a stand-scale model ensured a greater number of returns per sample, thus preserving canopy structure characteristics. This result again underscores the efficacy of stand-scale models in maintaining high landscape-level FACS prediction accuracy under reduced point density conditions. However, low point density may discourage local-level FACS pattern analysis. The partitioned model focusing solely on clearcut and patch cut samples produced a 59% RMSE (17.82 Mg/ha), indicating inaccurate prediction accuracy for young forests regenerating from severe disturbances. While the averaging effect inherent in the random forest algorithm contributed to model variance, the primary source of uncertainty likely came from inadequate non-ground returns, resulting in biased height metrics extraction. Future research may delve deeper into the mechanisms of return classification to address this issue. In summary, this study underscores the potential of the 3DEP LiDAR system as a promising tool for evaluating aboveground carbon stocks at a landscape scale in Indiana's hardwood forests.

2.7 References

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CHAPTER 3. CONCLUSIONS

The first study revealed that temperate hardwood forests at MMYSF store approximately 115.3 ± 2.1 Mg/ha of FACS, with a mean annual increase of 1.2 ± 0.4 Mg/ha. By comparing plot-level FACS estimates across various silvicultural treatment sites, it was found that intermediate treatments, such as single-tree selection (UEA) and prescribed burning (EA), had minimal impact on FACS over 13 years. In contrast, intensive management activities, including clearcutting (EA) and patch cutting (UEA), resulted in a significant reduction of FACS to 14.2 ± 6.2 Mg/ha to 39.1 ± 7.1 Mg/ha, which accounted for only 30 - 37% of their pre-treatment levels. The untreated forest at MMYSF is a typical example of middle-aged, secondary forests in the eastern US, showcasing rapid carbon accumulation in comparison to old-growth forests. Future research endeavors may use long-term inventory data to evaluate chronic carbon stock responses.

The second study emphasized the feasibility of using 3DEP LiDAR system as a valuable tool for evaluating landscape-level FACS in Indiana's hardwood forests. It was found that increasing the spatial scale from plots to stands significantly reduced the RMSE from 25.43 Mg/ha (26%) to 16.74 Mg/ha (20%). This reduction was attributed to reduced co-registration error and boundary effects at the stand scale than compared to the plot scale. Furthermore, stand-scale models also showed consistent prediction accuracy even with lower LiDAR point density, highlighting their effectiveness in maintaining high landscape-level FACS prediction accuracy. However, a partitioned model focusing solely on clearcut and patch cut samples exhibited a 59% RMSE (17.82 Mg/ha), indicating inaccurate prediction accuracy. This discrepancy was primarily due to insufficient non-ground returns and biased height metrics extraction. Future research should explore mechanisms of return classification to mitigate this issue and enhance the local-level prediction performance.

APPENDIX A. R CODE FOR FIELD-BASED FACS CALCULATION

```
library('tidyverse')#Tidyverse includes multiple packages such as dplyr, ggplot, ...
library('lattice')#This package is to create graphs/figures
library('ggfortify')#to enable autoplot function
library('GGally')#GGally is used to create correlation plots among multiple variables
library('psych')
library(ggpubr)
library(sjstats)
library(emmeans)
library(ggsci)

setwd("filepath")
ndf <- read.csv("./field_carbon/ref_carbon.csv")# Read in plot inventory data
ndf <- subset(ndf, Treatment!="EA Untreated") # remove EA untreated plots from analysis
bio08 <- subset(ndf, Year==2008)
bio13 <- subset(ndf, Year==2013)
bio17 <- subset(ndf, Year==2017)
bio21 <- subset(ndf, Year==2021)

#1. Let's first analyze biomass at each designated stands (pre-treatment, 2008)
table(bio08$Treatment)#shows how many plots are there for each stand
bio08$Treatment <- fct_relevel(bio08$Treatment,
                              'Control','Single-tree','Patch Cut','Clearcut','Shelterwood',"Burning")
md08 <- lm(data = bio08, agc~Treatment) #agc is the same as FACS
autoplot(md08)#Equal variance and normality assumptions are met for ANOVA. No need to
transform
summary(md08)#p=0.450, F (5,293)=0.949. No sig diff between treatments
anova_stats(md08, digits = 3)
means08 <- emmeans(md08, "Treatment")#Get means of each stand
means08# displays the means
```

```
#pairs(means08)# estimates the P values of the pairwise comparisons
means08_df <- data.frame(means08) #Make it a dataframe
```

#2. 2013 data

```
bio13$Treatment <- fct_relevel(bio13$Treatment,
                              'Control','Single-tree','Patch Cut','Clearcut','Shelterwood','Burning")
md13 <- lm(data = bio13,agc~Treatment)
autoplot(md13)##Equal variance and normality assumptions are met for ANOVA.
summary(md13)#p<0.001, F(5,358) = 51.33
anova_stats(md13,digits = 3)
means13 <- emmeans(md13, "Treatment")#Get means of each stand
means13# displays the means
pairs(means13)# S3 method for class "emmGrid" (p14)
```

#3. 2017 data

```
bio17$Treatment <- fct_relevel(bio17$Treatment,
                              'Control','Single-tree','Patch Cut','Clearcut','Shelterwood','Burning")
md17 <- lm(data = bio17,agc~Treatment)
autoplot(md17)#Equal variance and normality assumptions are met for ANOVA.
summary(md17)#p<0.001, F(5,544) = 71.03
anova_stats(md17,digits = 3)
means17 <- emmeans(md17, "Treatment")#Get means of each stand
means17# displays the means
pairs(means17)
```

#4. 2021 data

```
bio21$Treatment <- fct_relevel(bio21$Treatment,
                              'Control','Single-tree','Patch Cut','Clearcut','Shelterwood','Burning")
md21 <- lm(data = bio21,agc~Treatment)
autoplot(md21)#Equal variance and normality assumptions are met for ANOVA.
summary(md21)#p<0.001, F(5,544) = 62.19
```

```
anova_stats(md21,digits = 3)
means21 <- emmeans(md21, "Treatment")#Get means of each stand
means21# displays the means
pairs(means21)
```

APPENDIX B. R CODE FOR RANDOM FOREST MODEL ESTABLISHMENT

Note: This appendix only includes necessary R codes for demonstrating the process of using
##recursive feature elimination, building the basic stand-level random forest model for FACS
##prediction, and using random forest model to create FACS prediction maps.

```
library('tidyverse')
library('lattice')
library(terra)
library('sf')
library('ggfortify')
library('psych')
library(ggpubr)
library(sjstats)
library(exactextractr)
library(randomForest)
library(caret)
setwd("filepath")

## Protocols to build basic stand-level model
#1.Read in ref 2013
ref13 <- read.csv("stand_hee13_Ref_agb_v2.csv")#read in ref agc(stand-level FACS)
sum_stand13 <- read.csv("hee13_px_to_stand_varExtraction.csv")#read in lidar variables
#Now we need to assign years to sum_stand13 to create equal "pin" to match two datasets
sum_stand13$Year <- 2013
sum_stand13$stand <- paste(sum_stand13$Mgmt_Area,"_",
                           sum_stand13$Year,sep = "")
#Now merge with ref13 file
sum_stand13 <- merge(sum_stand13,ref13)
sum_stand13$Mgmt_Area <- NULL
```

```

sum_stand13$Year <- NULL
sum_stand13 <- relocate(sum_stand13,Treatment,.after = stand)
sum_stand13 <- relocate(sum_stand13,agc,.after = Treatment)

#2. Read in ref 2017
ref17 <- read.csv("stand_hee17_Ref_agb_v2.csv")#read in ref agc
sum_stand17 <- read.csv("hee17_px_to_stand_varExtraction.csv")#read in lidar variables
#Now we need to assign years to sum_stand17 to create equal "pin" to match two datasets
sum_stand17$Year <- 2017
sum_stand17$stand <- paste(sum_stand17$Mgmt_Area,"_",
                           sum_stand17$Year,sep = "")
#Now merge with ref17 file
sum_stand17 <- merge(sum_stand17,ref17)
sum_stand17$Mgmt_Area <- NULL
sum_stand17$Year <- NULL
sum_stand17 <- relocate(sum_stand17,Treatment,.after = stand)
sum_stand17 <- relocate(sum_stand17,agc,.after = Treatment)
ov_ht17 <- subset(sum_stand17,Treatment=="Clearcut" | Treatment == "Patch Cut")
median(ov_ht17$CHM_mean)

#3. Merge 2013, 2017 files and perform random forest.
#Merge 2013,2017 data
hee_stand <- rbind(sum_stand13,sum_stand17)
aggregate(hee_stand$agc,by=list(hee_stand$Treatment),FUN="mean")
aggregate(hee_stand$agc,by=list(hee_stand$Treatment),FUN="sd")
#All stands
all_stands <- hee_stand[,c(2:41)]#Extract biomass and metric columns
#Partition independent variables and dependent variables
x <- all_stands%>%
  select(-c("agc","Treatment"))%>%
  as.data.frame()
y <- all_stands$agc

```

```

set.seed(123)
inTrain <- createDataPartition(y,p=0.8,list=F)[,1]
x_train <- x[inTrain,]
x_test <- x[-inTrain,]
y_train <- y[inTrain]
y_test <- y[-inTrain]

#Set rfe control
control <- rfeControl(functions = rfFuncs,
                      method = "repeatedcv",
                      repeats = 5,
                      number = 10)
result_rfe <- rfe(x=x_train,y=y_train,sizes = c(1:39),rfeControl = control)
result_rfe
predictors(result_rfe)# Read rfe results

#Now keep suggested variables
df_train <- x_train[,c("cc20m","CHM_max","cc30m","CHM_sd","cc15m","QMH","ht_kurt")]
df_test <- x_test[,c("cc20m","CHM_max","cc30m","CHM_sd","cc15m","QMH","ht_kurt")]

set.seed(123)
rd_all <- randomForest(x=df_train,
                      y=y_train,
                      xtest=df_test,
                      ytest = y_test,
                      importance=T,ntree=500)# var tried at each split =12
rd_all#r2=0.79 (0.78)
sqrt(280.1501)#16.74
sqrt(280.1501)/mean(y_train)#20.38%
sqrt(281.54)#16.78
sqrt(281.54)/mean(y_test)#19.9%

```

```

#Create a carbon map
fx <- rbind(df_train,df_test) #combine the training set and testing sets of predictor variables
fy <- c(y_train,y_test) #combine the training set and testing sets of response variable (FACS)
#Train the 2013 map
cmap_v1 <- rast("cmap_v1.tif") #Read in raster layers (variables)
names(cmap_v1)
rd_final <- randomForest(x=fx,y=fy,importance = T,ntree = 500)
carbon_hee13 <-
terra::predict(cmap_v1,rd_final,filename="carbon_hee13_standscale.tif",na.rm=T,overwrite=T)
carbon_hee13#15-134
rm(carbon_hee13)
rm(cmap_v1)
#Train the 2017 map
cmap_v2 <- rast("cmap_v2.tif")
carbon_hee17 <-
terra::predict(cmap_v2,rd_final,filename="carbon_hee17_standscale.tif",na.rm=T,overwrite=T)
carbon_hee17#10-134
rm(carbon_hee17)
rm(cmap_v2)

```